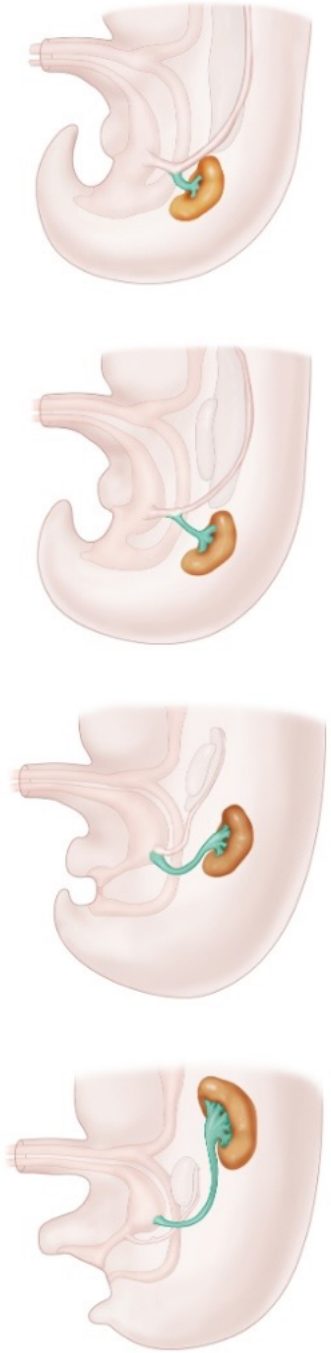
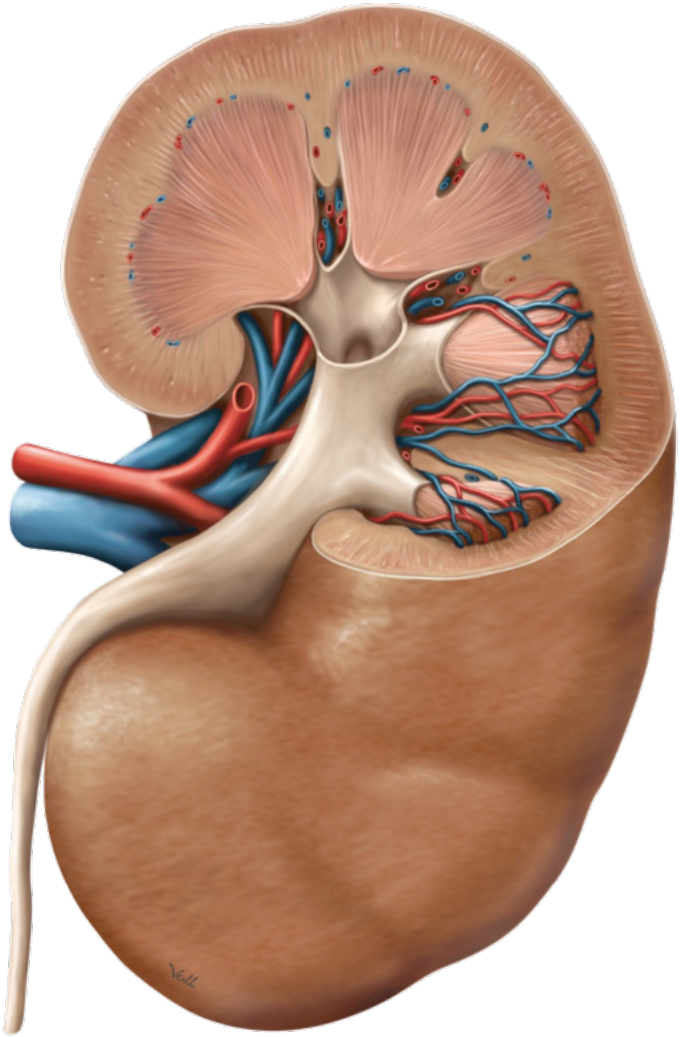


Ascent of the metanephros



Right Kidney, Posterior View

Renal Functional Anatomy & Development

Ellis Locke, Ph.D.
Idaho College of Osteopathic Medicine

Session Objectives

1. Review the gross anatomy of the adult kidney including the renal capsule, cortex, medulla, and structures within the renal sinus.
2. Identify the different segments of the uriniferous tubule and relate each segment to its location and function within the kidney.
3. Explain the circulation of the kidney, including the components of renal microvasculature.
4. Compare and contrast the origin, differentiation, and fate of the three embryonic kidneys, with emphasis on the development of the metanephros.
5. Describe developmental disruptions that can occur during development, migration, and rotation of the metanephros as they relate to congenital anomalies.

Session Readings

Junqueira's Basic Histology, 18th Ed. Mescher, 2027.
Chapter 19: The Urinary System – Introduction, Kidneys, and Blood Circulation

Before We Are Born, 10th Ed. Moore, Persaud, and Torchia. 2020.
Chapter 13: Urogenital system (section: Development of the Urinary System)

Online Resources

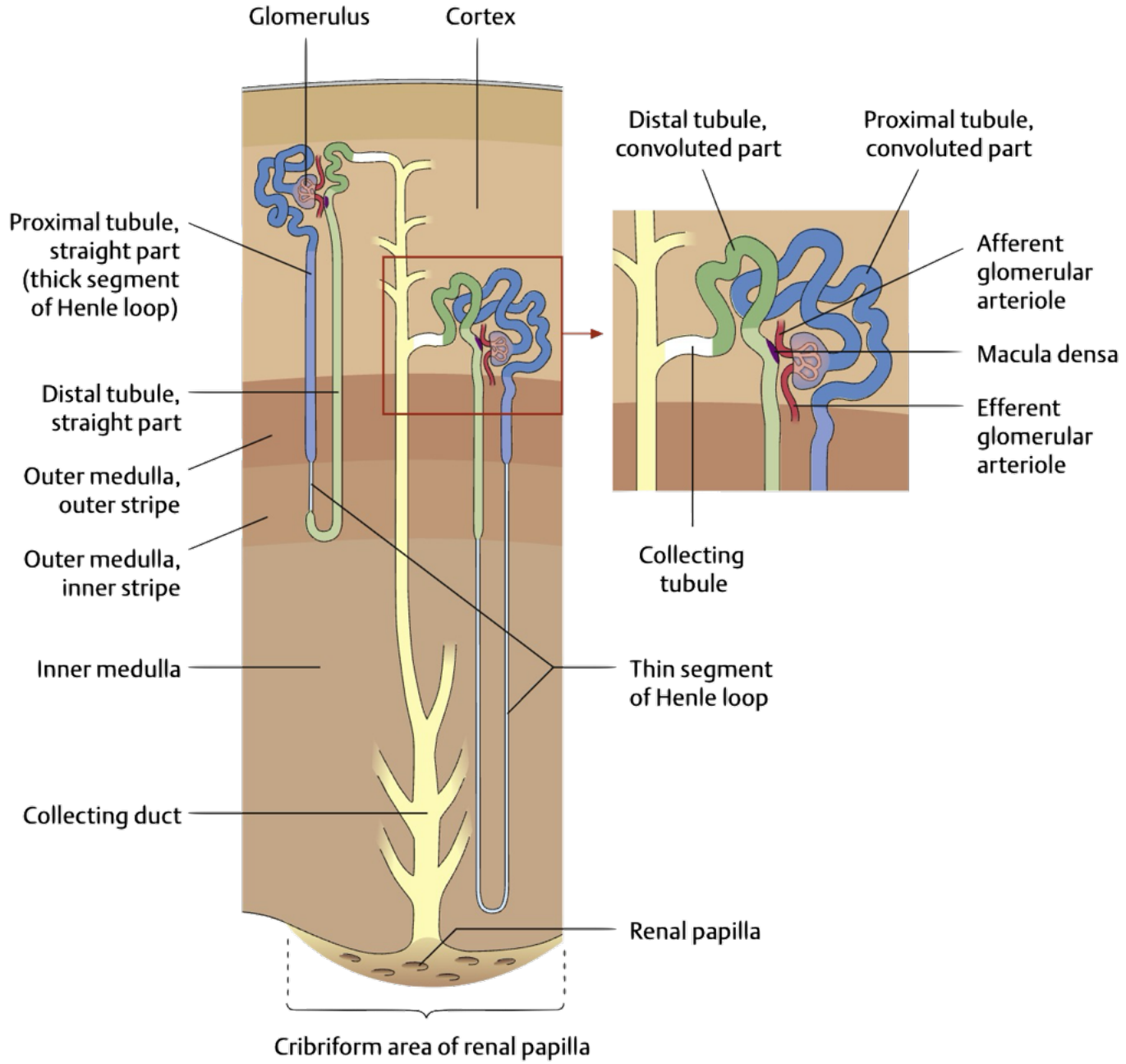
- ICOM Histology Site – RENL: Renal Function Anatomy & Development
  - Histology Guide – Urinary System: Kidney

Image Sources

Before We Are Born, 10th Ed. Moore, Persaid, and Torchia. 2020.
Langman's Medical Embryology, 14th Ed. Sadler, T.W. 2020.
Junqueira's Basic Histology, 16th Ed. Mescher, 2021.
Histology: A Text and Atlas, 8th Ed. Pawlina & Ross, 2020.
Wheater's Functional Histology, 6th Ed. Young et al, 2014.
Thieme Atlas of Anatomy, 4th Ed. Gilroy, McPherson, & Ross. 2020

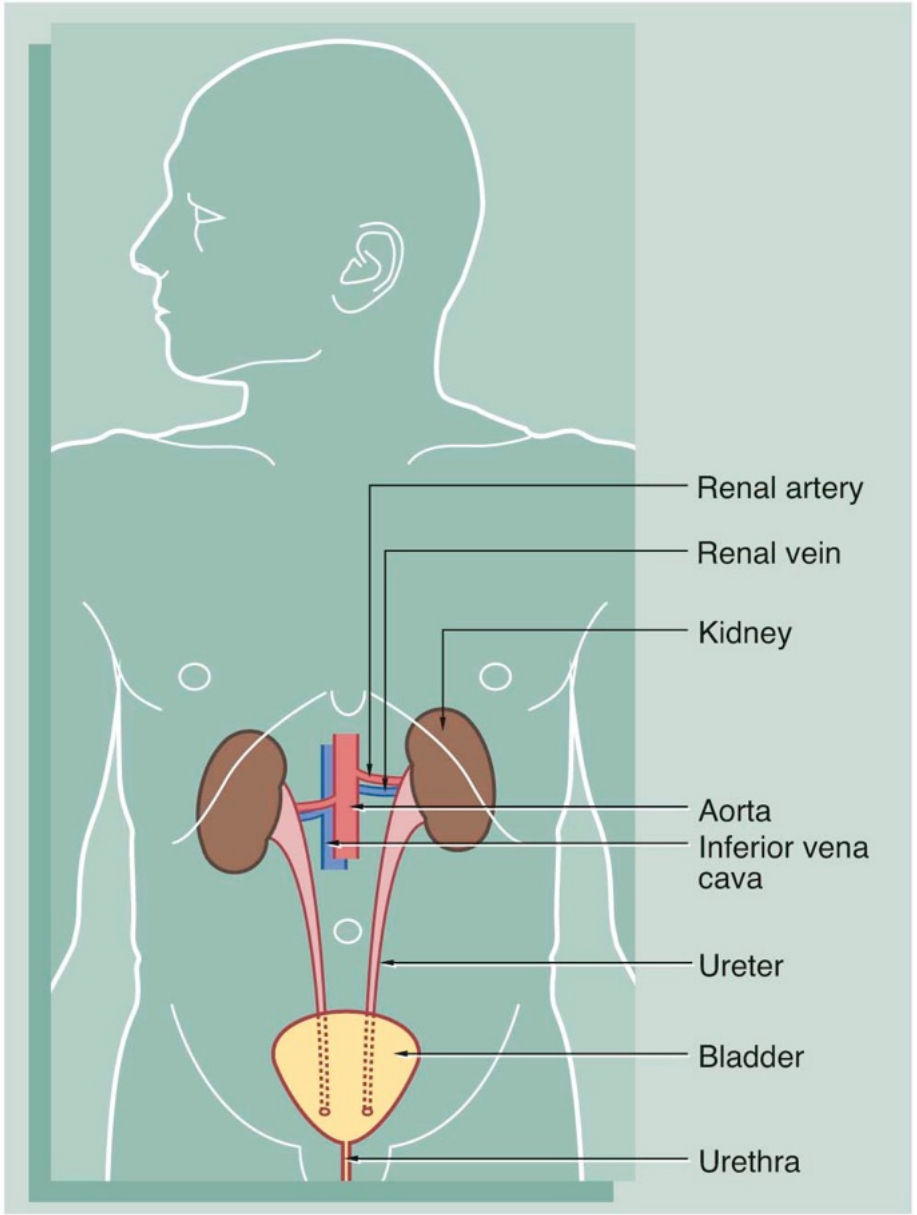
Session Outline

- Overview of Renal Functions
- Gross Anatomy Review
- Microanatomy of the Kidney
- Renal Circulation
- Renal Embryology Review

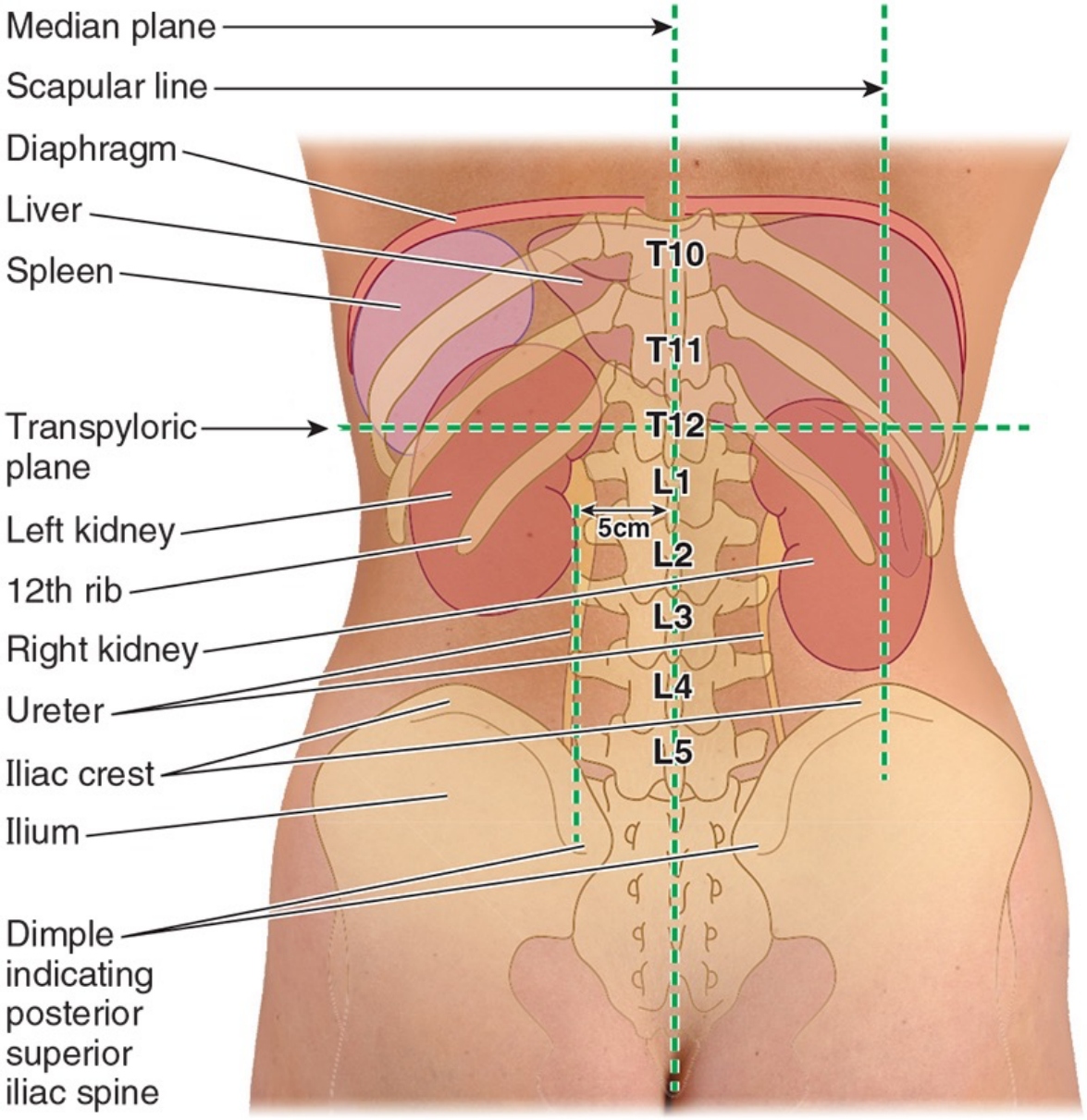


General Renal Functions

- **Maintain homeostasis:** osmoregulation and acid-base balance
- **Waste Excretion:** metabolic waste, excess water & electrolytes, and bioactive substances removed from the blood & excreted into urine
- **Endocrine functions:** synthesis of **renin** (part of RAAS control of blood pressure) and **erythropoietin** (stimulates erythrocyte production with blood O2 is low)
- **Vitamin D conversion:** hydroxylation of steroid prohormone vitamin D into active form calcitriol



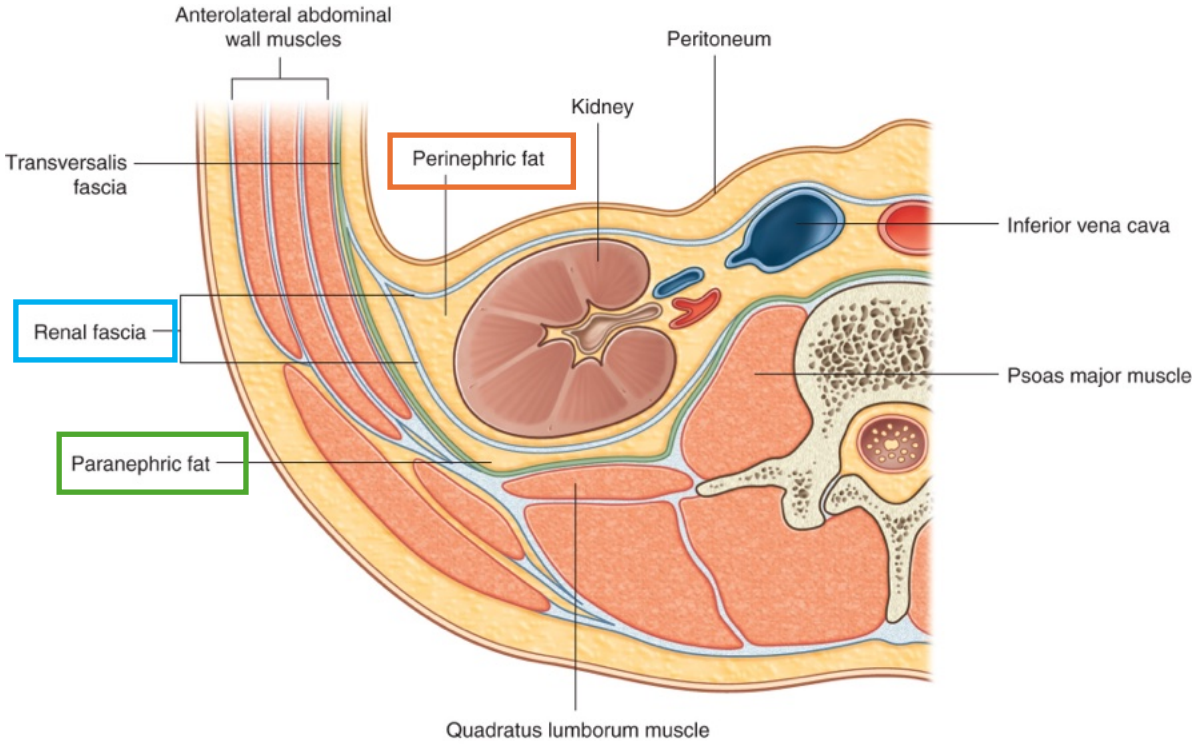
Kidneys



Posterior view

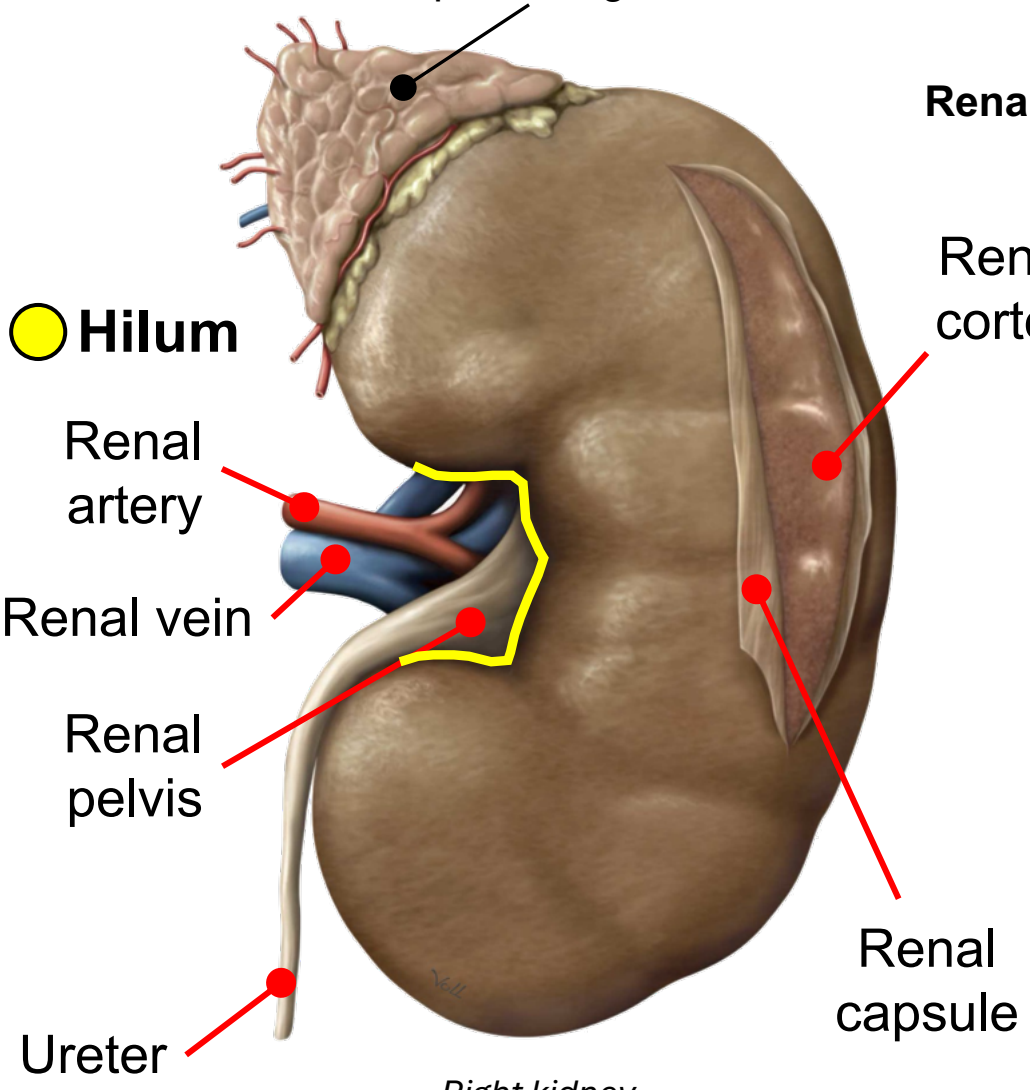
Primarily retroperitoneal organs

- T12-L3 vertebral levels
- Right kidney slightly lower than left
- Surrounded by 3 layers of fat & fascia (paranephric fat, renal fascia, perinephric fat)

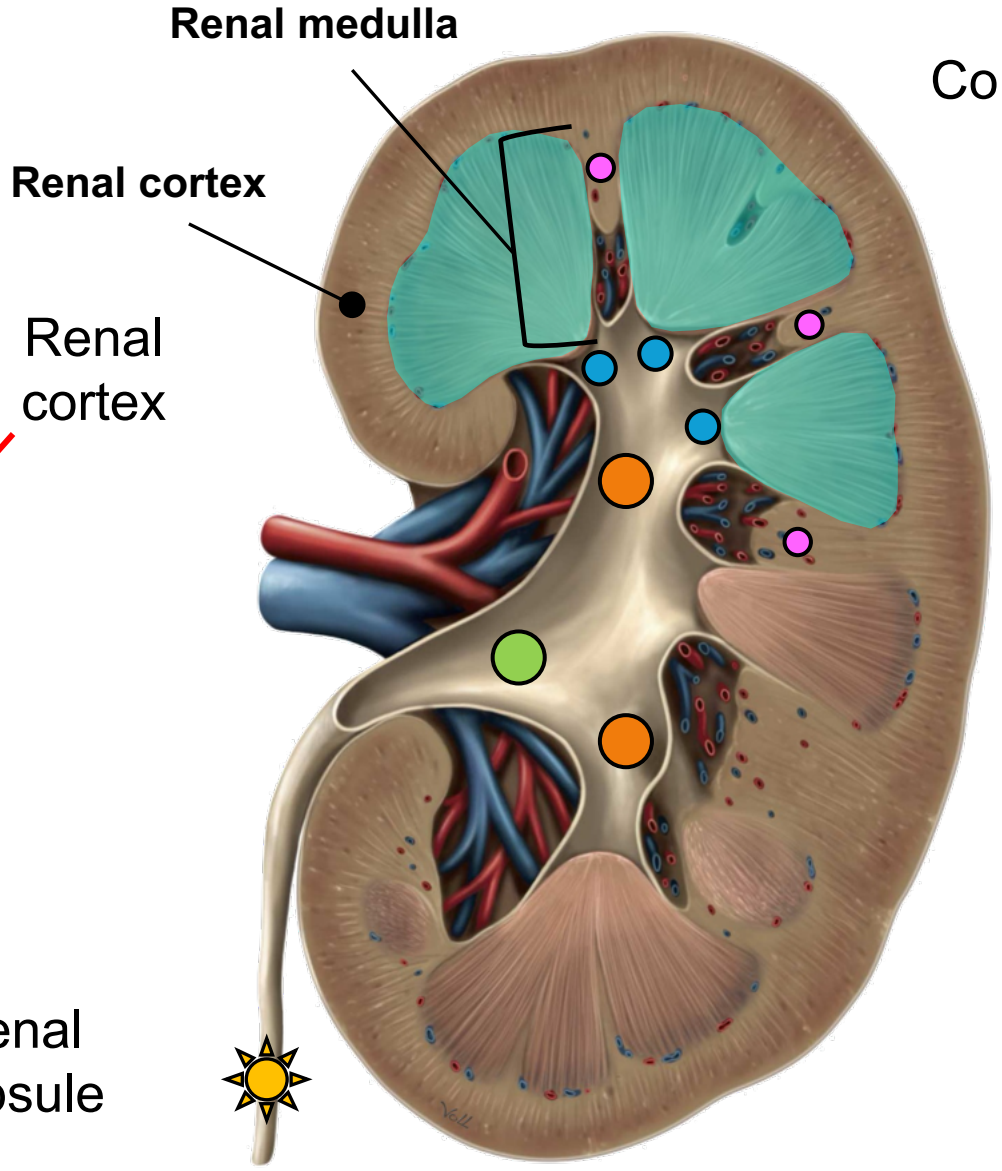


Transverse section

Kidneys: External & Internal Gross Anatomy



Right kidney, Posterior view



Midsagittal section of right kidney Posterior view

Cortex extends internally as
renal columns

renal pyramids

↓

minor calyx

↓

major calyx
(receives 2-3 minor calices)

↓

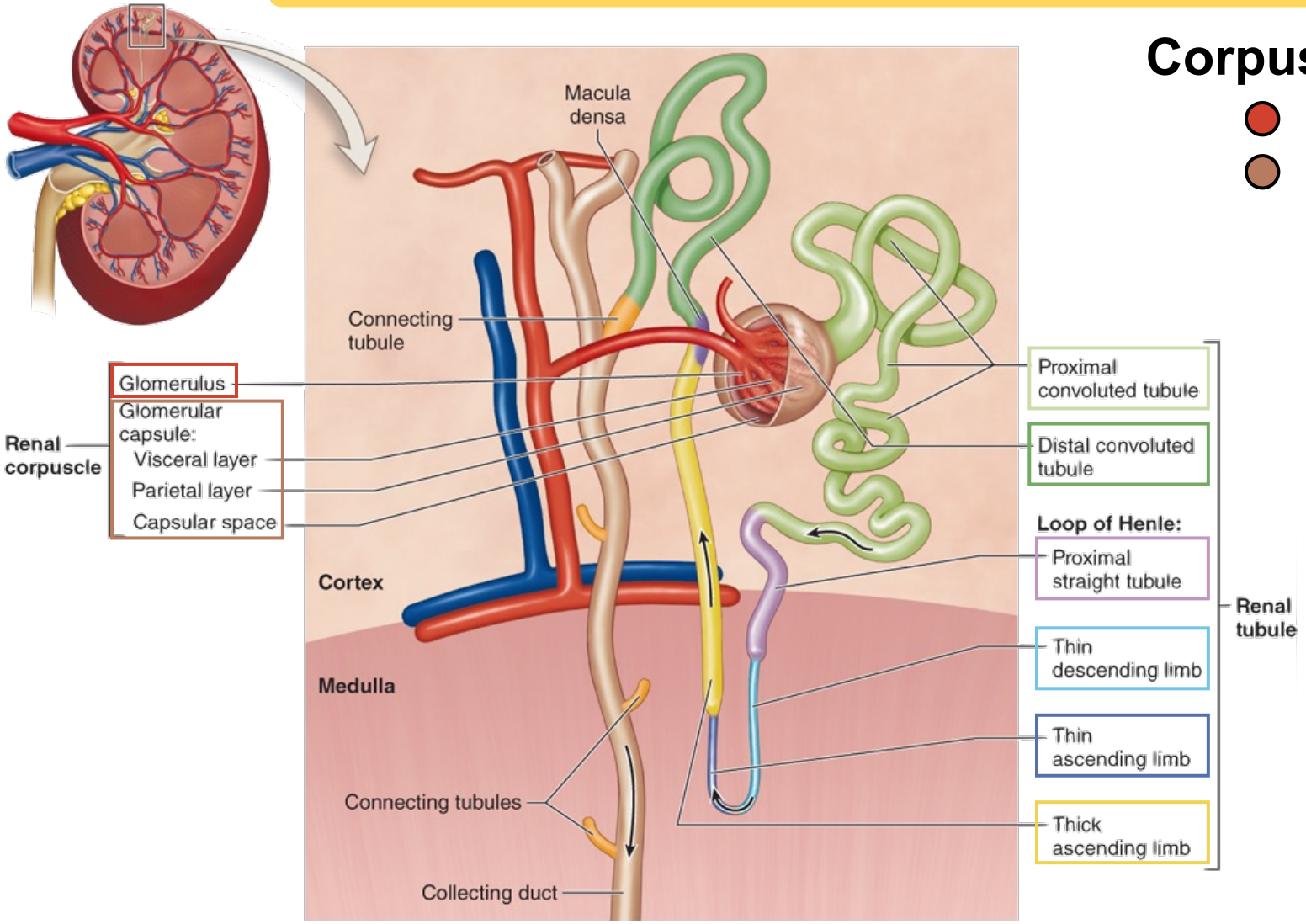
renal pelvis
(receives 2-3 major calices)

↓

ureter

Excretory System

Nephron: functional unit of the kidneys (~1.25 million per kidney)



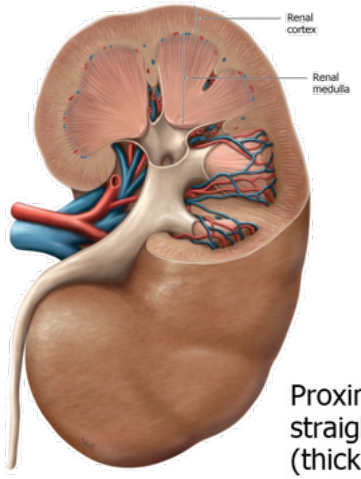
Corpuscles: produce filtrate from blood

- glomerulus (tuft of capillaries)
- glomerular (Bowman's) capsule

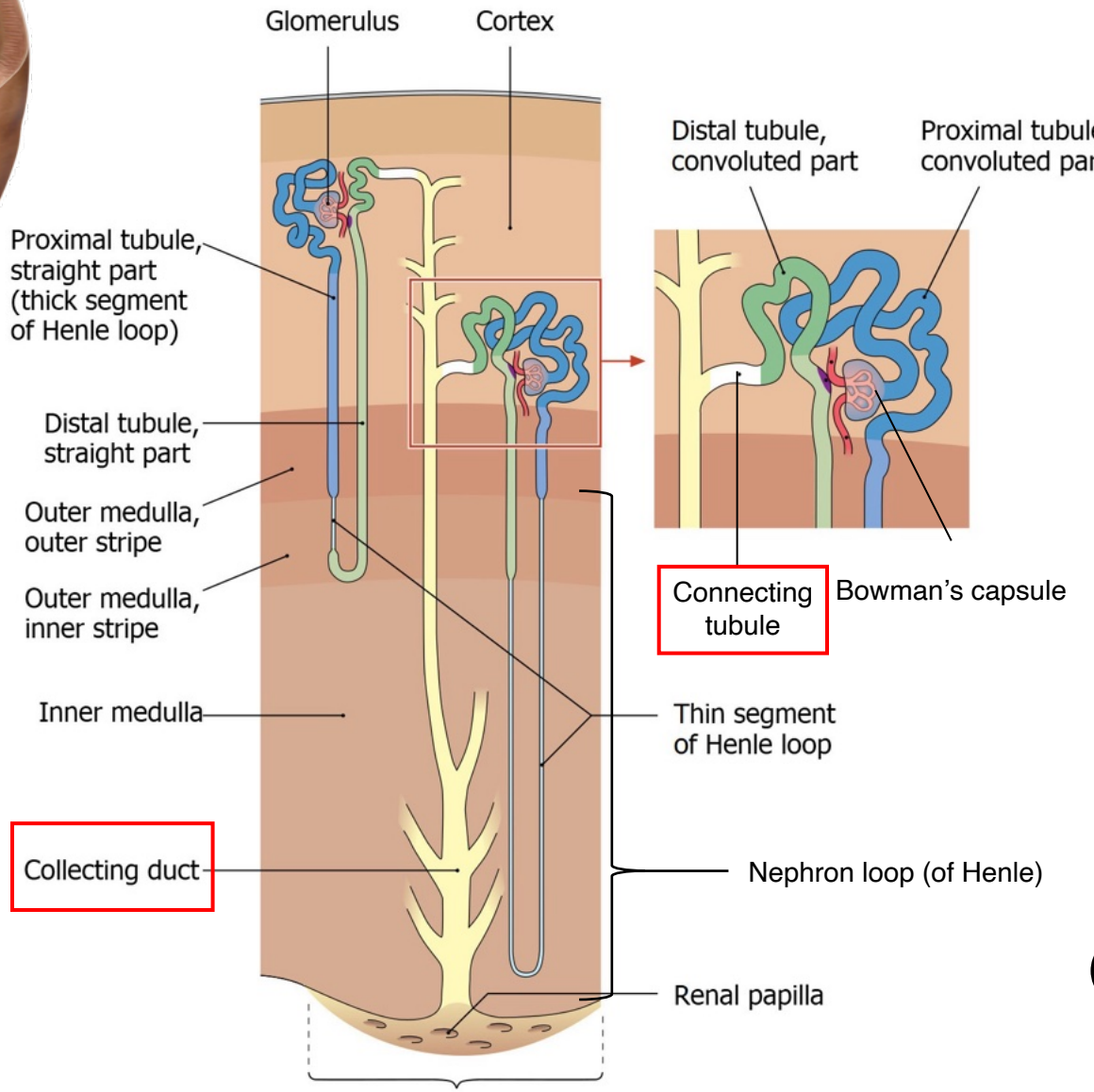
Tubules: process filtrate before it enters the collecting system

- **Proximal convoluted tubule**
- **Nephron loop (of Henle):**
  - **Thick descending limb** / proximal straight tubule
  - **Thin descending limb**
  - **Thin ascending limb**
  - **Thick ascending limb** / distal straight tubule (with **macula densa**)
- **Distal convoluted tubule**

Collecting System



Posterior view



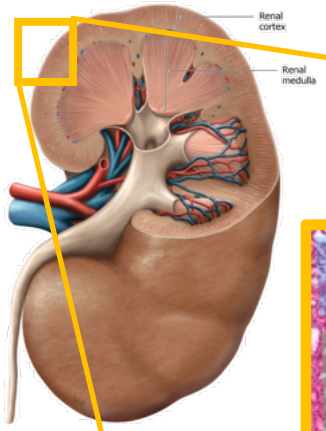
Distal convoluted tubule → **connecting tubule** (start of the collecting system) → **collecting duct**

Each collecting duct gathers filtrate from multiple nephrons & is the final site of tubular fluid modification

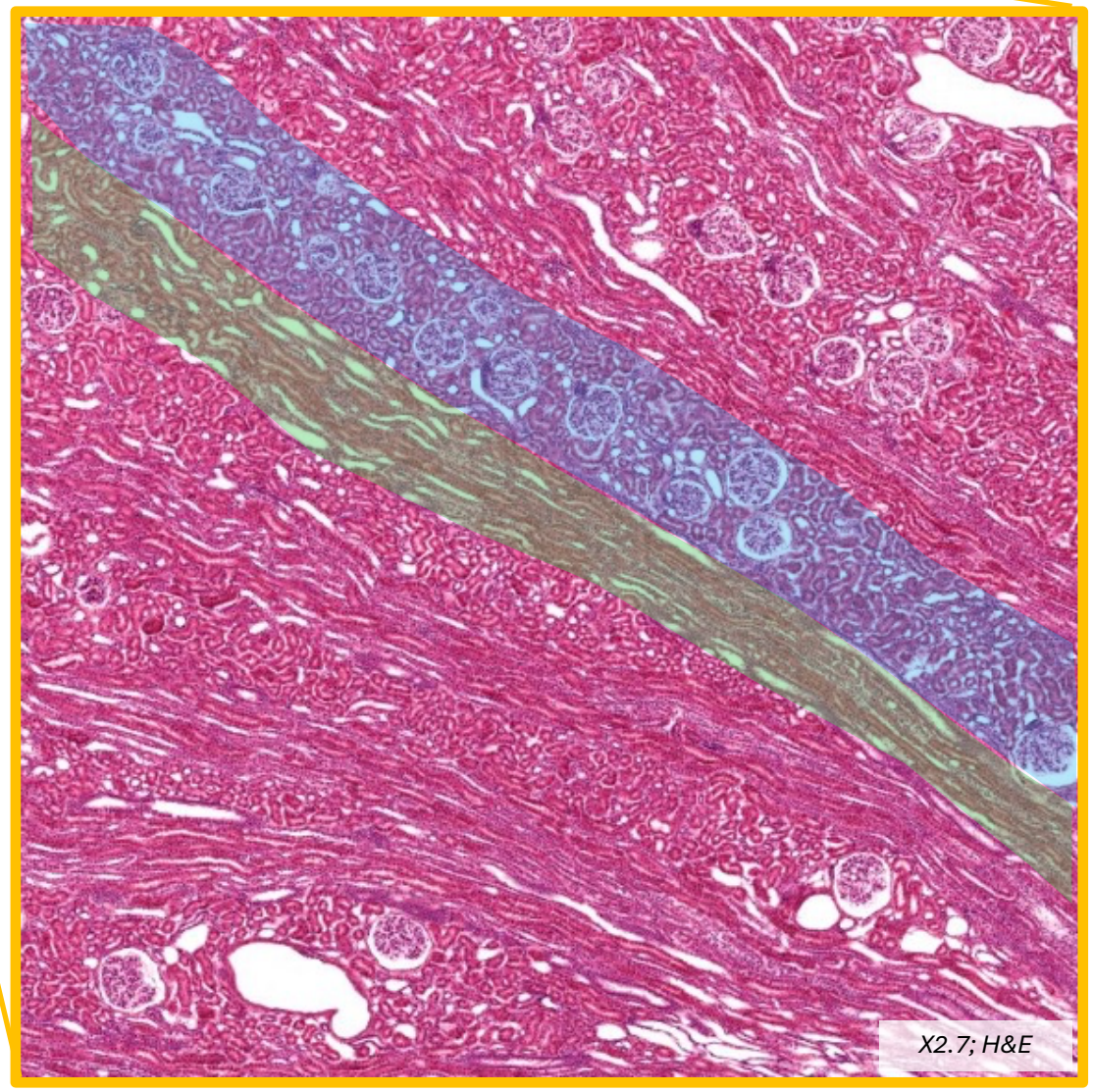
Multiple collecting ducts merge & drain urine through the **papillary ducts** at the area cribrosa of the **renal papilla** (located at the apex of each renal pyramid)

'Uriniferous tubule' refers to a nephron + its associated collecting duct

Renal Cortex



Posterior view



HistologyGuide

Contains renal corpuscles, associated tubules, and extensive vasculature

Medullary ray: cortical **collecting duct** & associated **straight tubules** (PST & DST)

Cortical labyrinth: located between medullary rays, containing **renal corpuscles** and their associated **convoluted tubules** (PCT & DCT)

Renal Medulla

Medullary tissue forms **renal pyramids**
(base at ● corticomedullary junction; apex at ● papilla)

Contains **straight tubules**, continuations of the **collecting ducts**, and the **vasa recta** (a specialized capillary network running parallel to tubules)

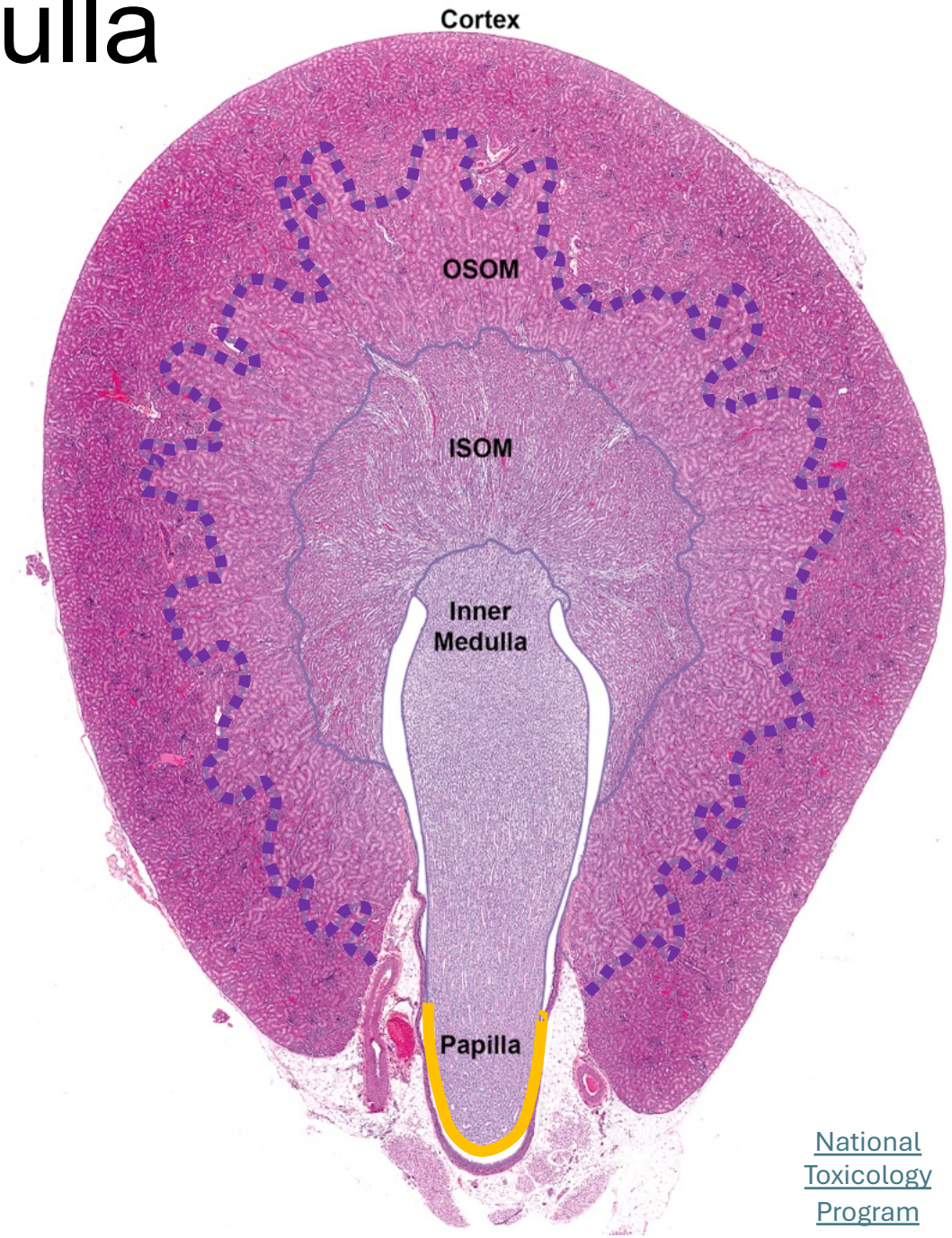
Outer Medulla

- **OSOM:** Outer stripe of the outer medulla
- **ISOM:** Inner stripe of the outer medulla

&

Inner Medulla

Medullary interstitium has an **osmotic gradient**: solute concentration increases from the corticomedullary junction to the renal papilla



National Toxicology Program

Tubules of the Medulla

OSOM

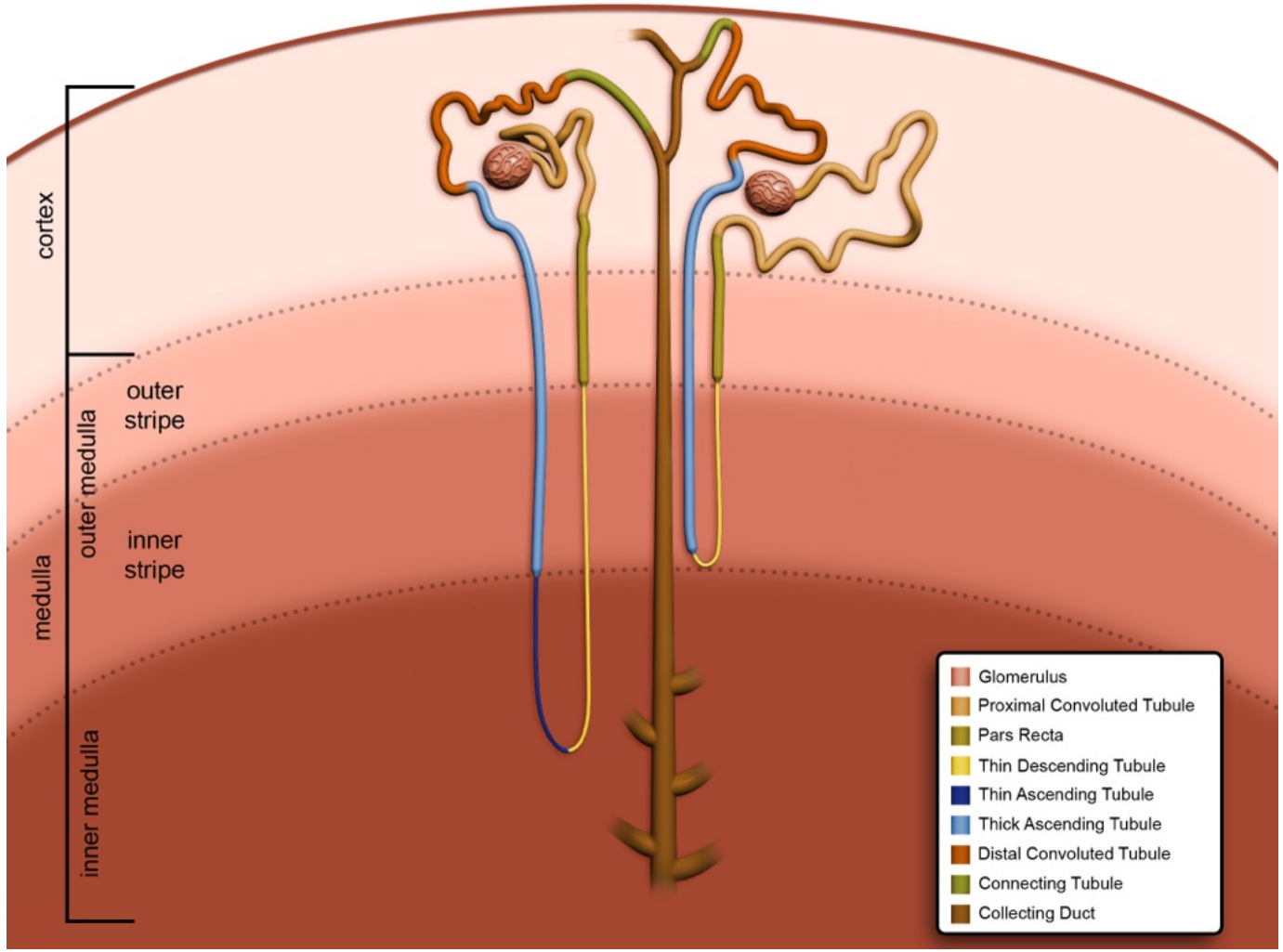
- **Thick limbs** of the nephron loop
- Continuation of the **collecting duct**

ISOM

- **Thin descending limb** and **thick ascending limb** of the nephron loop
- Continuation of the **collecting duct**

Inner medulla

- **Thin limbs** of the nephron loop
- Continuation of the **collecting ducts** → converge to form **papillary ducts**



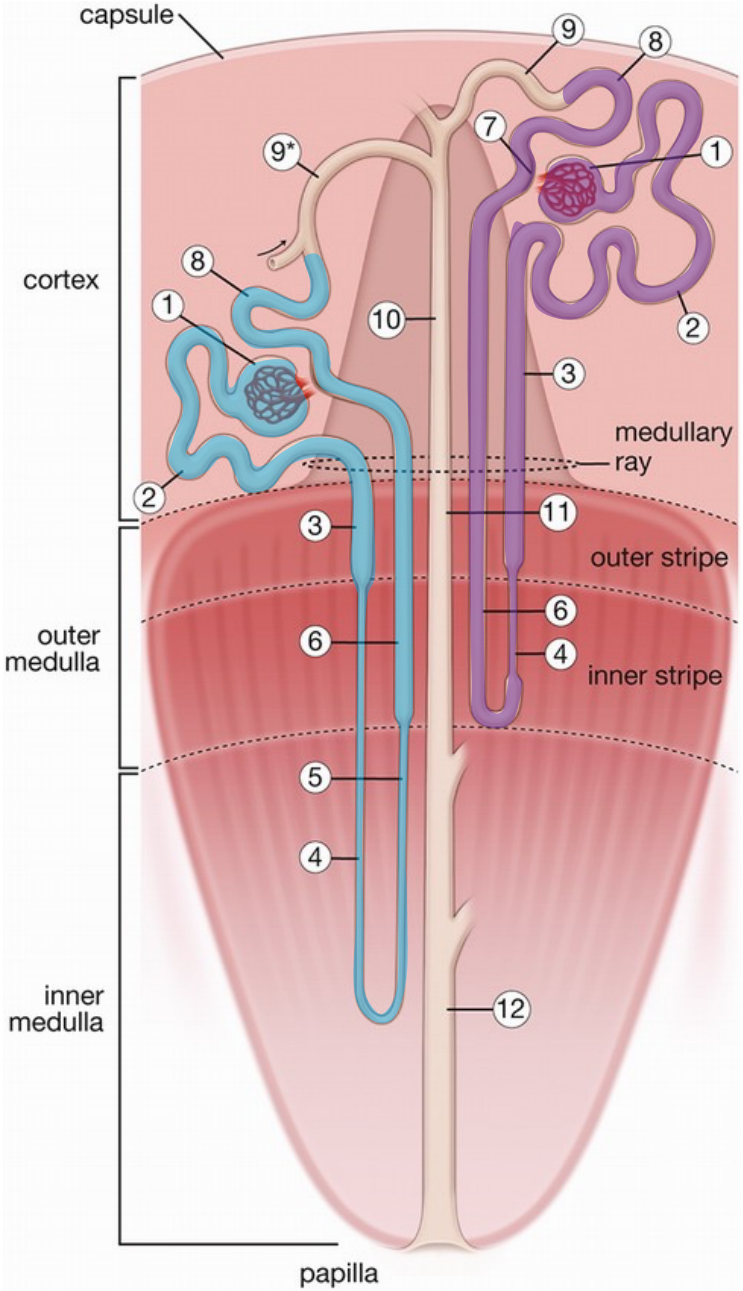
National Toxicology Program

Nephron Classification

Nephrons are classified based on **location of the renal corpuscle** and the **length / depth** of the nephron loop

Juxtamedullary nephrons:
8 – 10% of nephrons

- Nephron loops extend into the inner medulla → produce most concentrated tubular fluid



Cortical nephrons:
90 – 92% of nephrons

- **Subcapsular nephrons** have corpuscles superficial in the cortex; short nephron loops extend only into the outer medulla → less concentrated tubular fluid

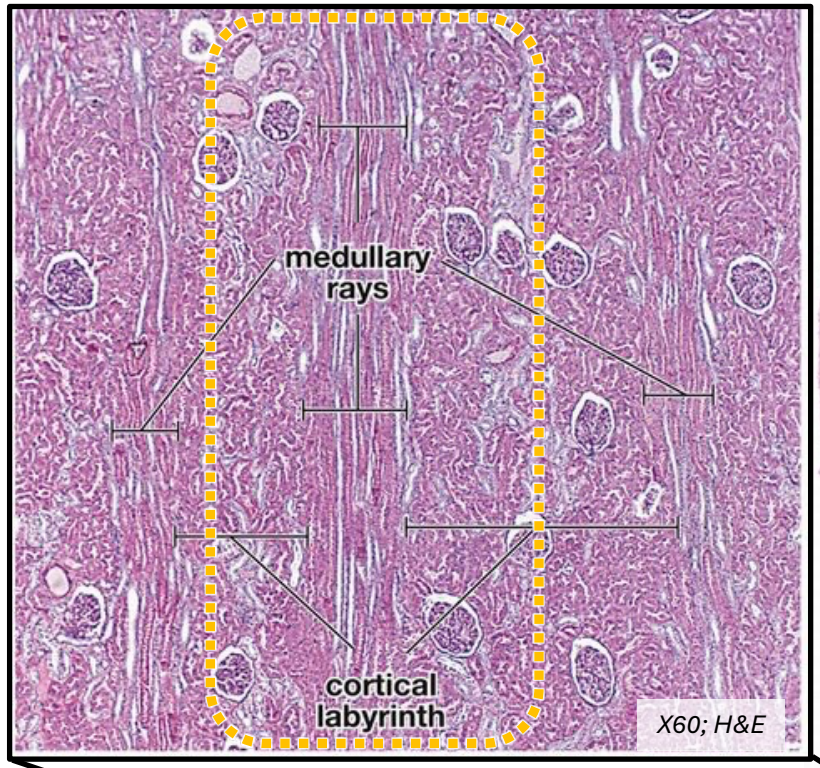
- **Midcortical nephrons:** have corpuscles in the mid cortex; nephron loops intermediate in length between subcapsular & juxtamedullary nephrons
(not pictured)

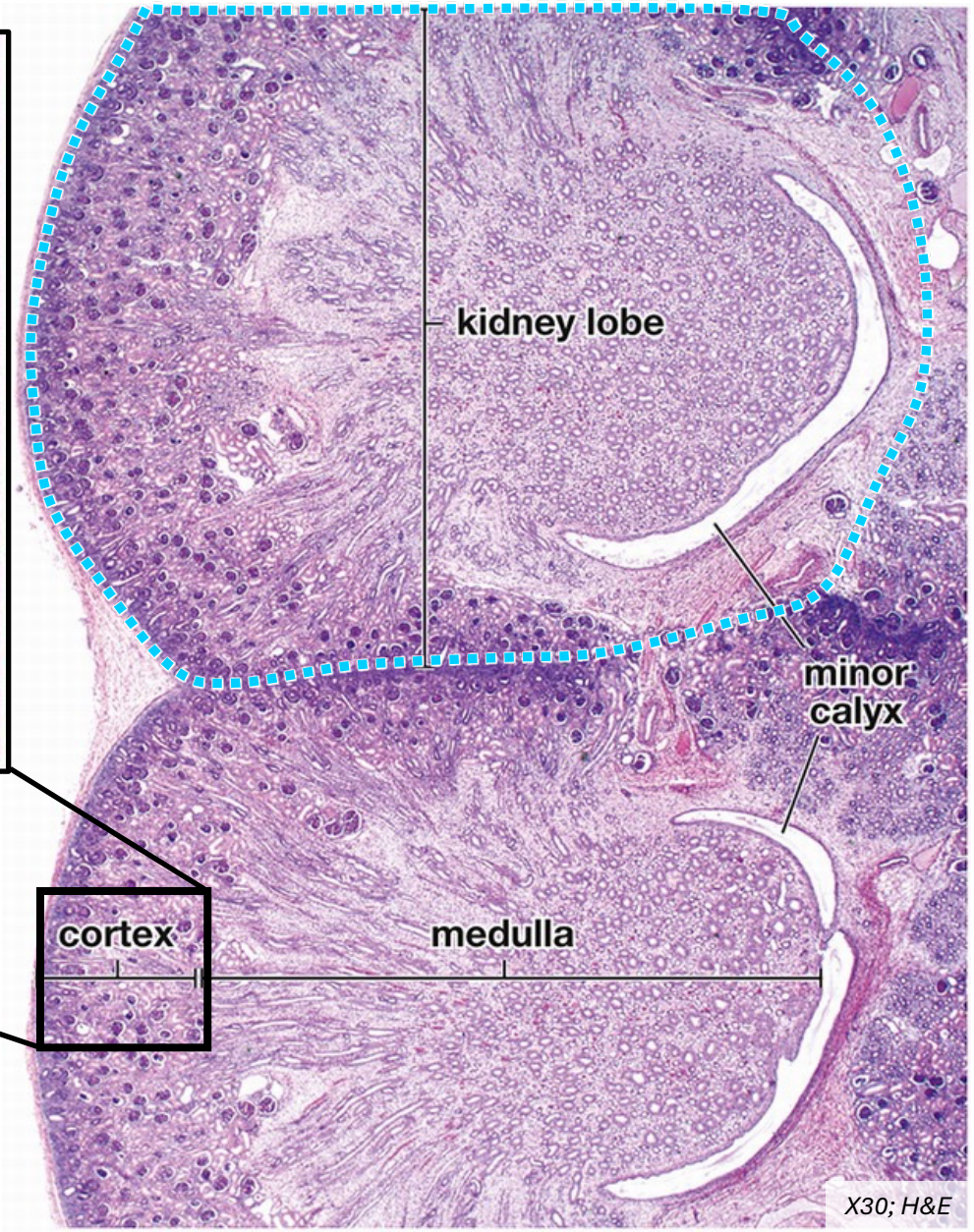
Lobes & Lobules

Renal lobe

(8-15 per human kidney)

- Pyramid, associated minor calyx, ½ of adjacent columns & overlying cortex
- Cortex of each lobe contains multiple lobules



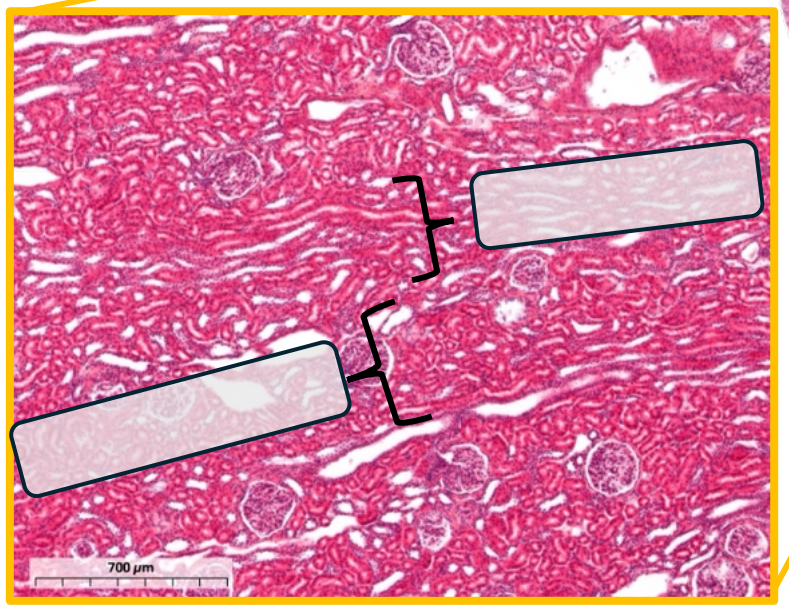


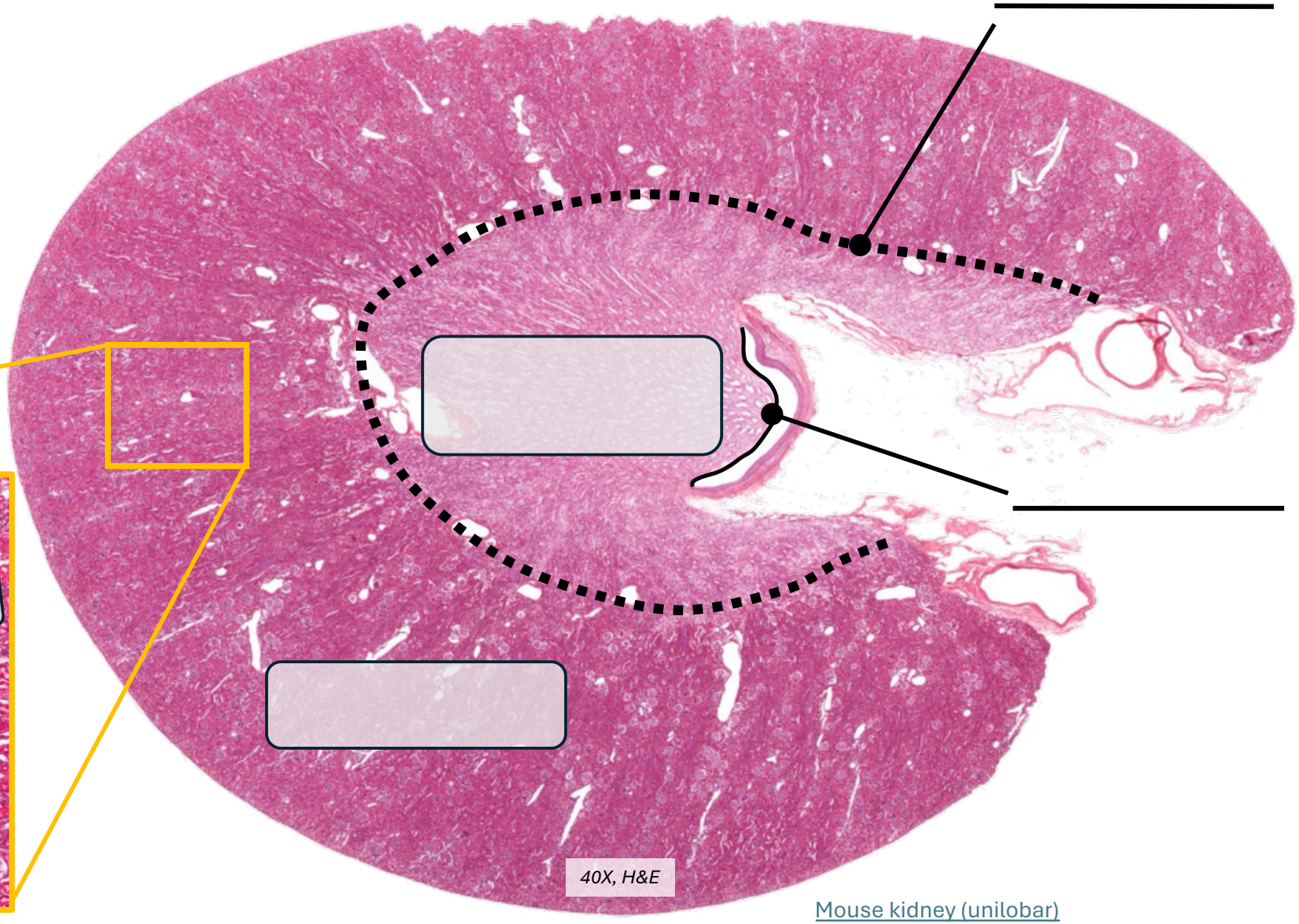
Human fetal kidney

Renal (cortical) lobule

- Cortical collecting duct + associated tubules of the nephrons that it drains
- Central **medullary ray** & roughly ½ of the **cortical labyrinth** tissue on either side
- Boundaries between lobules are not obviously demarcated

Pause & Recall

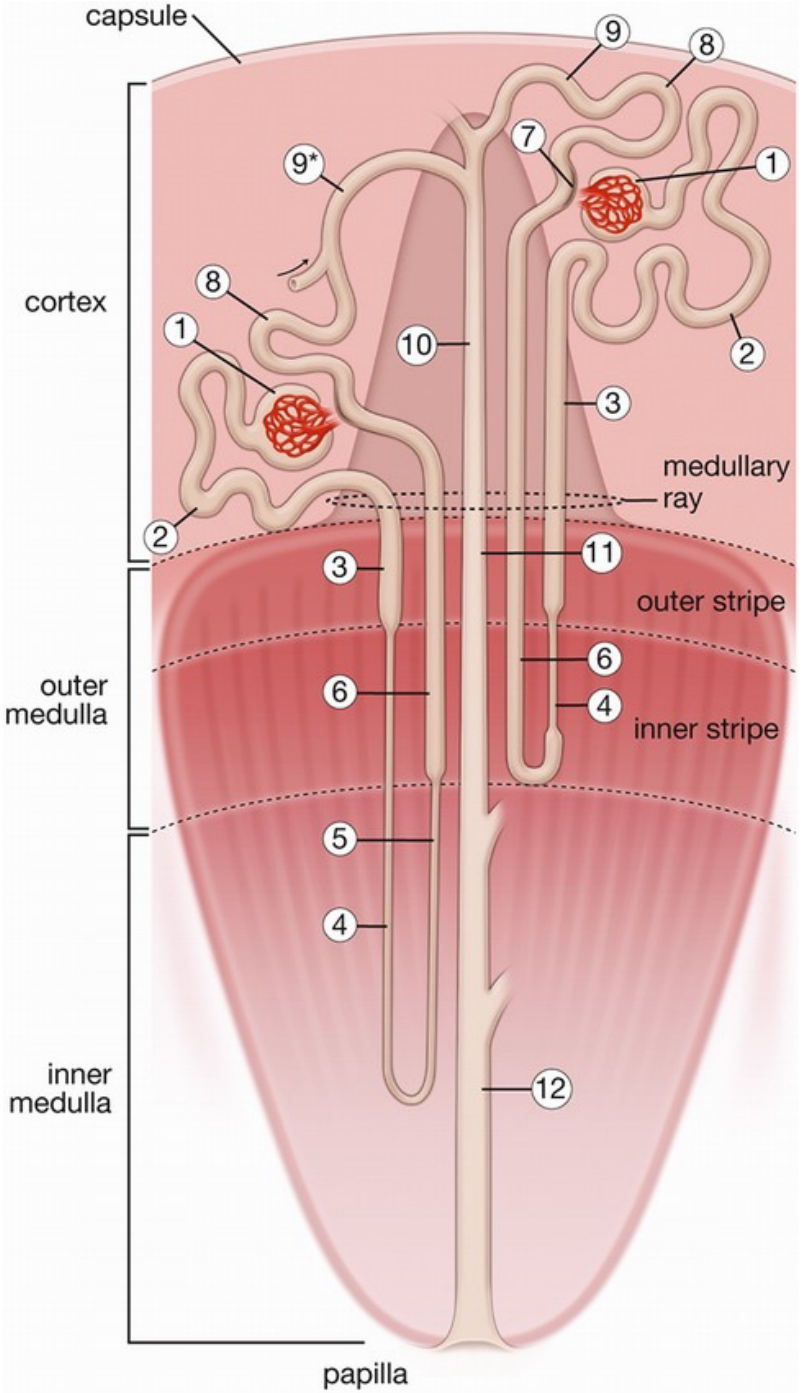




Mouse kidney (unilobar)
HistologyGuide

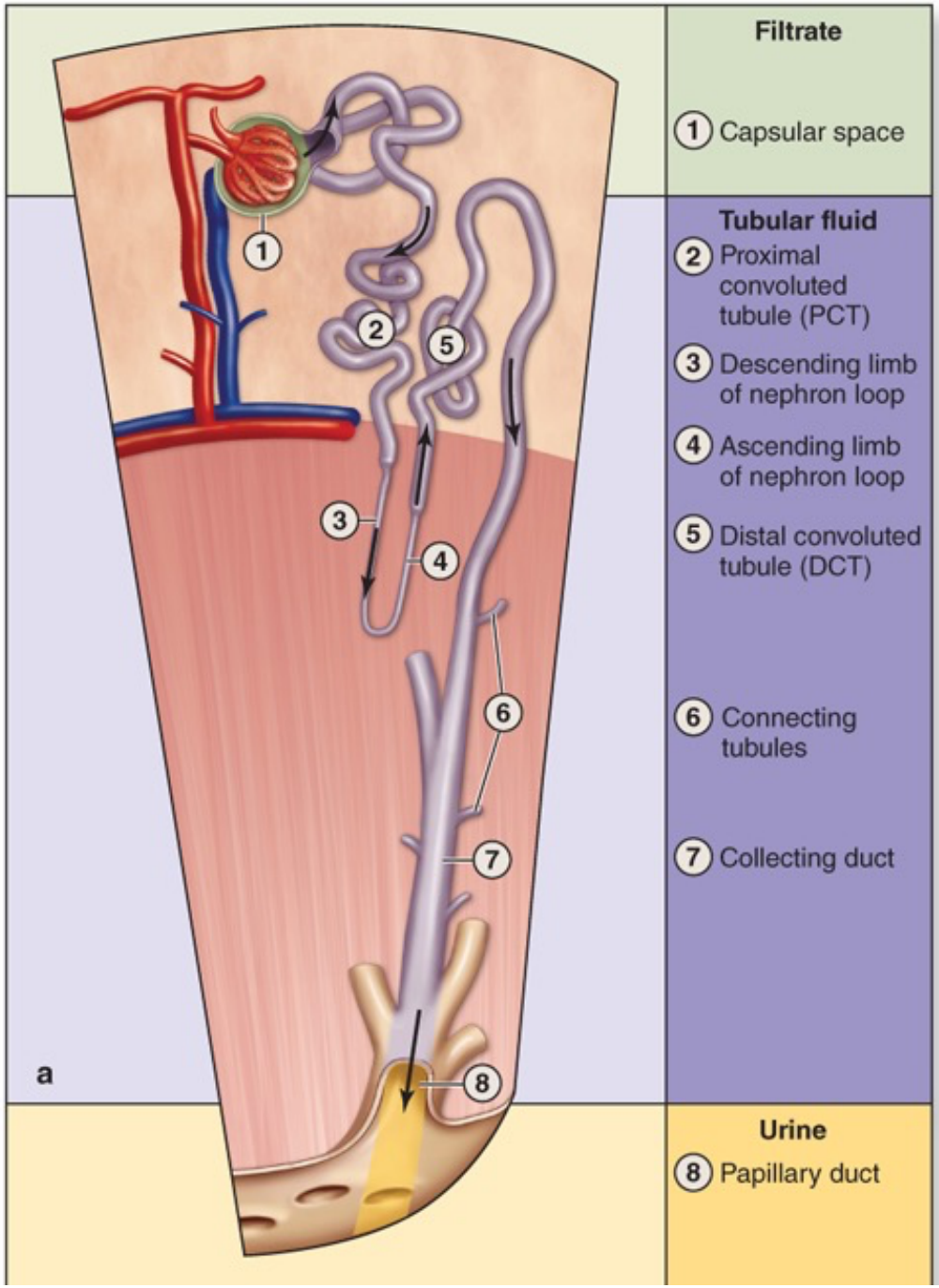
Pause & Recall

Identify the numbered components of the uriniferous tubule



1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____

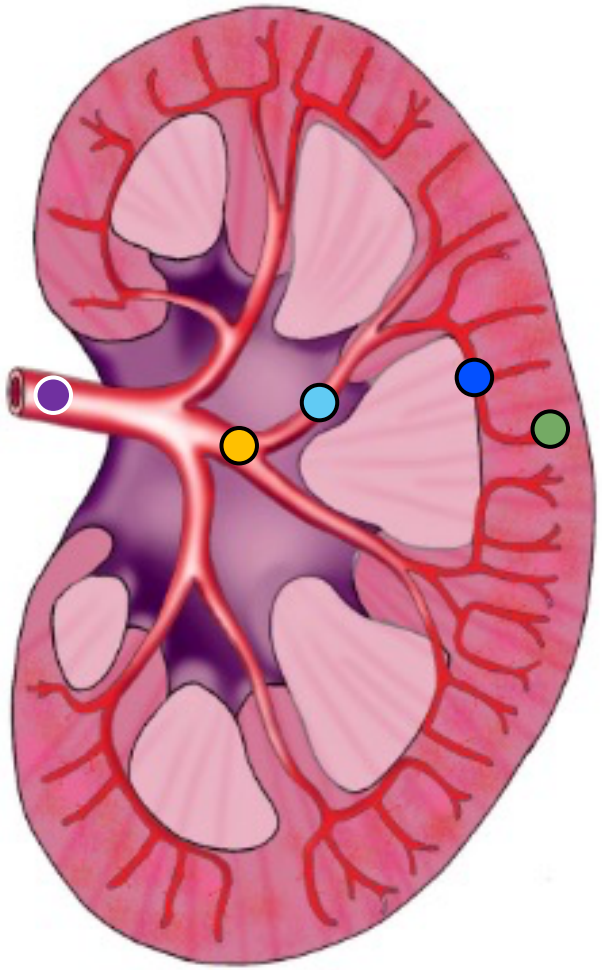
10. – 12. _____

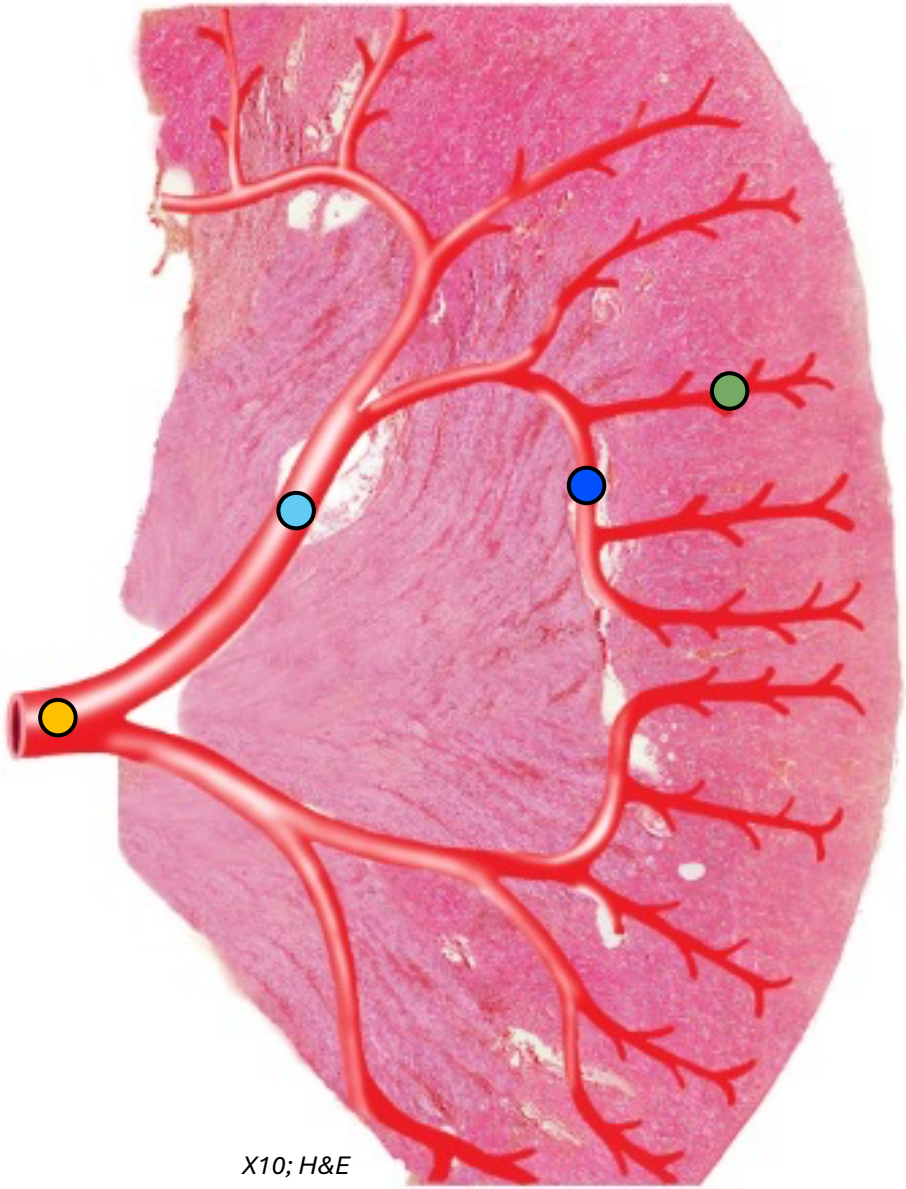


Uriniferous Tubule Functions

- **Glomerular filtration:** water & solutes from blood plasma exit the glomerulus and enter the lumen of the glomerular capsular space to form **ultrafiltrate**
- **Tubular reabsorption:** select substances in tubular fluid are transported across the epithelium into interstitial tissue & returned to the bloodstream
- **Tubular secretion:** substances in the interstitial tissue are transported across the epithelium & secreted into the tubular fluid
- Secretion and reabsorption mechanisms & targets (i.e., molecules being transported) vary along the length of the uriniferous tubule – *details in upcoming Renal Histology Session*

Renal Circulation





X10; H&E

Digital Histology

- Renal artery
- Segmental artery
- Interlobar artery
- Arcuate artery
- Interlobular artery
- Afferent arterioles *(to the glomerulus)*

Cortical Vasculature

A – afferent arteriole
G – glomerulus
I – interlobular arteries
PT – peritubular capillaries

Afferent arteriole

↓

Glomerulus

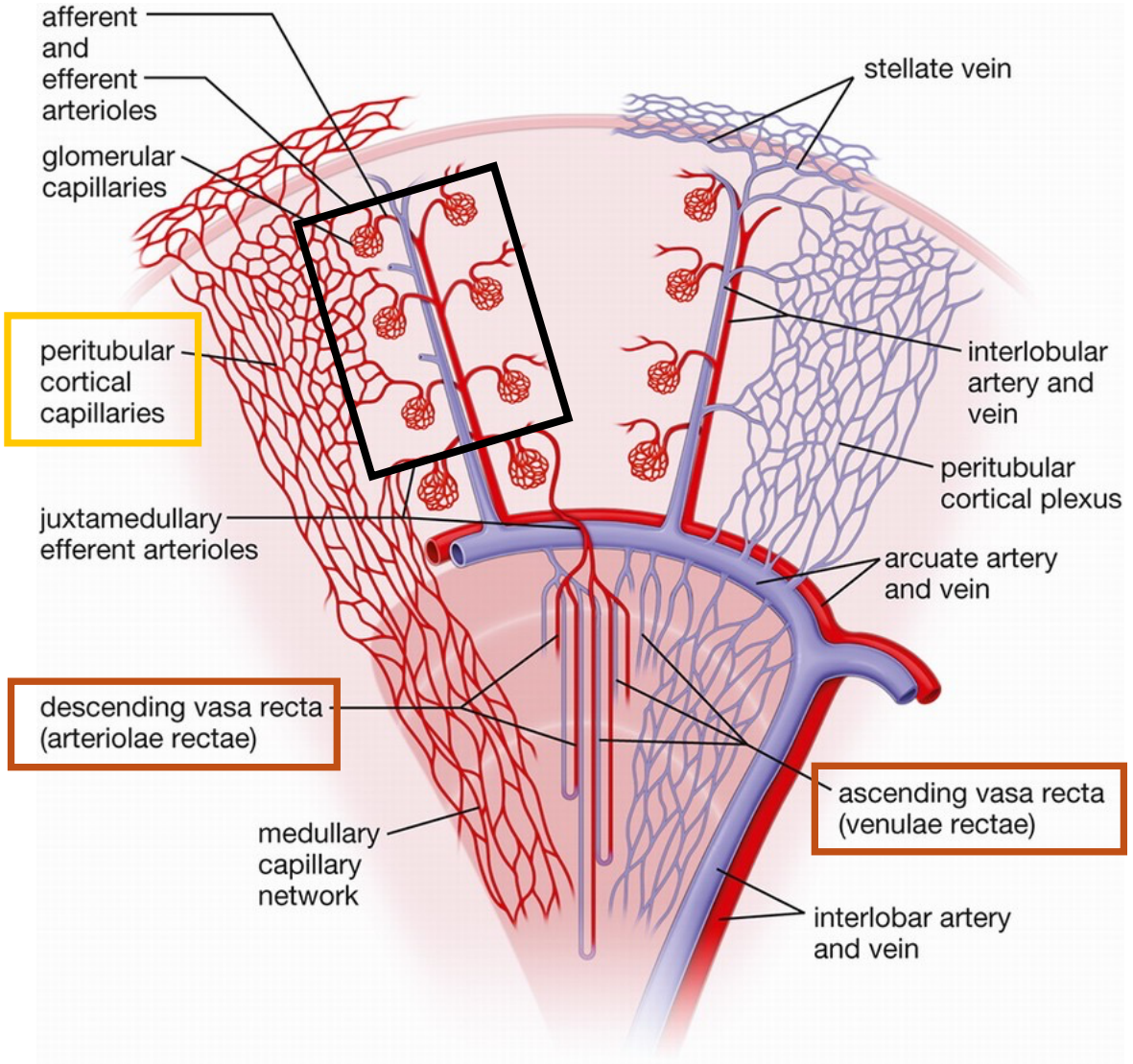
↓

Efferent arteriole

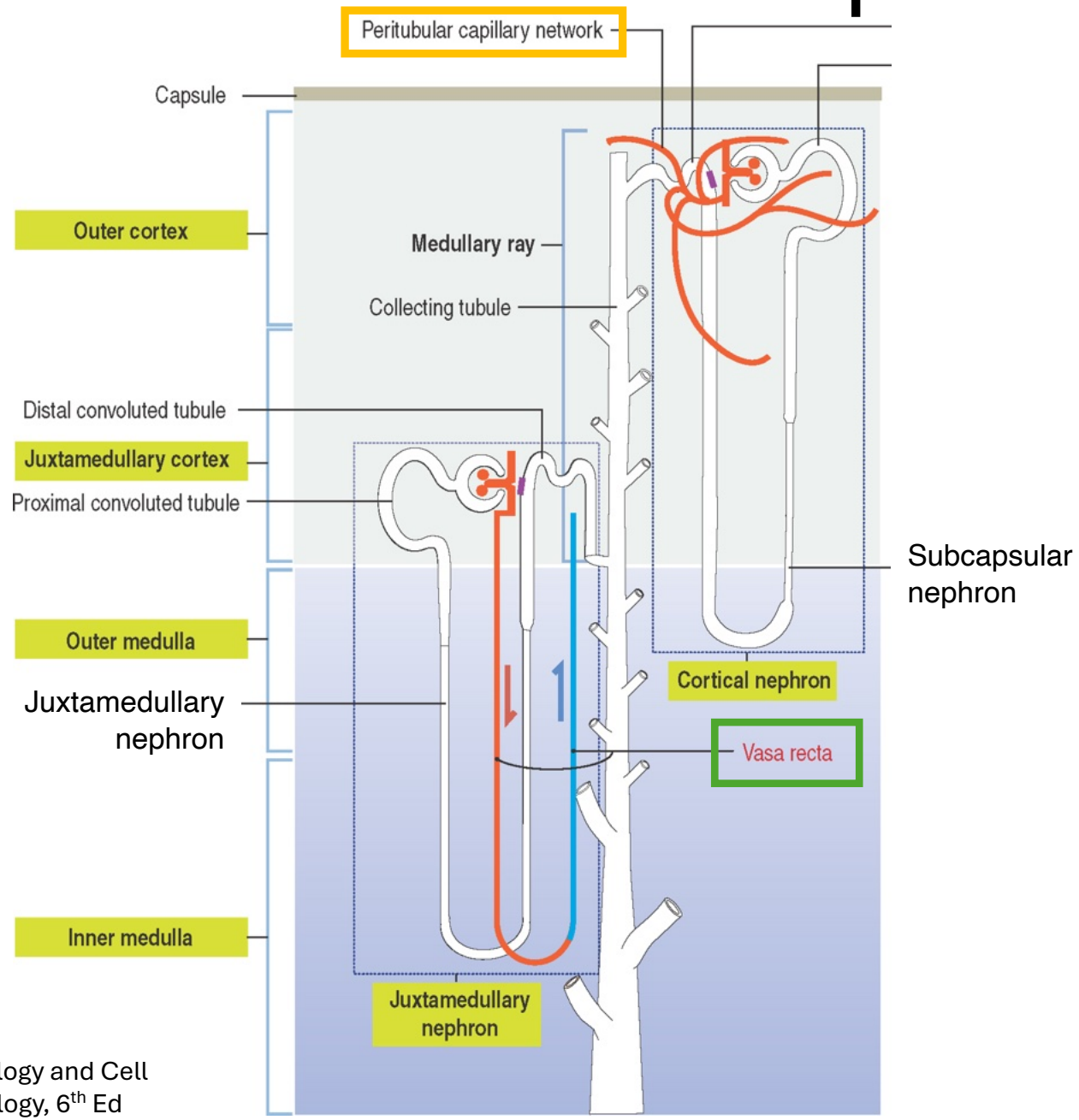
↓

- Peritubular capillaries
  or
- Vasa recta

X125; Arteries injected with carmine dye before fixation



Peritubular Capillaries & Vasa Recta



Histology and Cell Biology, 6th Ed

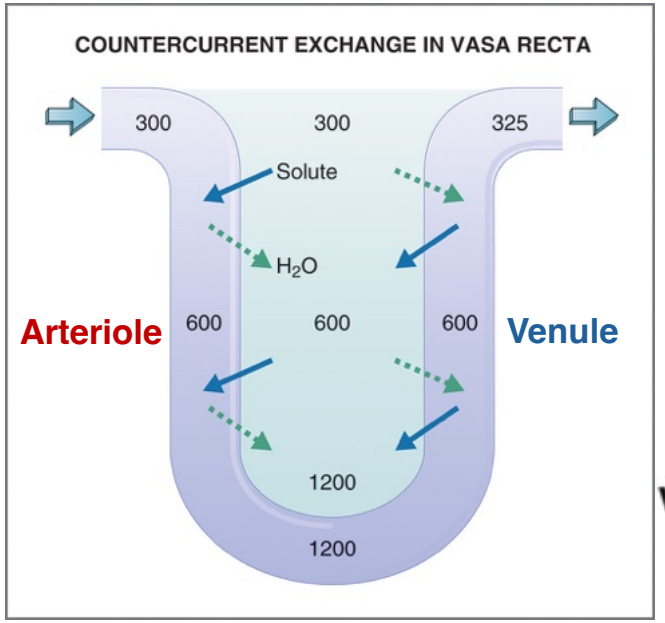
Peritubular capillaries

- Arise from efferent arterioles of **cortical nephrons** (subcapsular & midcortical)
- Limited to the renal cortex
- Associated with reabsorption & secretion of the **proximal & distal convoluted tubules**

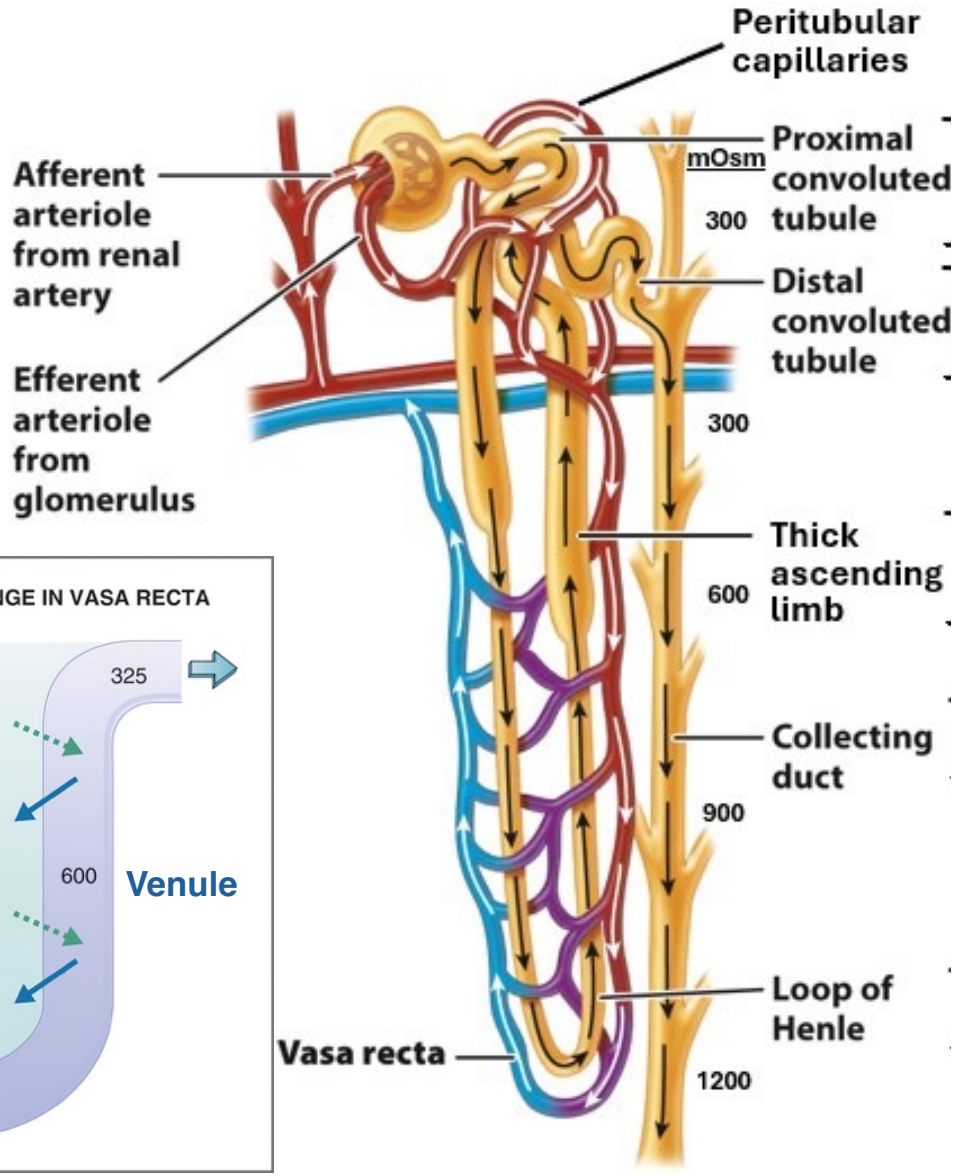
Vasa recta

- Arise from efferent arterioles of **juxtamedullary nephrons**
- Descend into the medulla & form **parallel loops** adjacent to the nephron loop
- Countercurrent flow with tubular fluid creates **countercurrent exchange system** & maintains osmotic gradient in the medulla

Countercurrent Exchange System



Costanzo Physiology, 8th ed.



Duke Histology

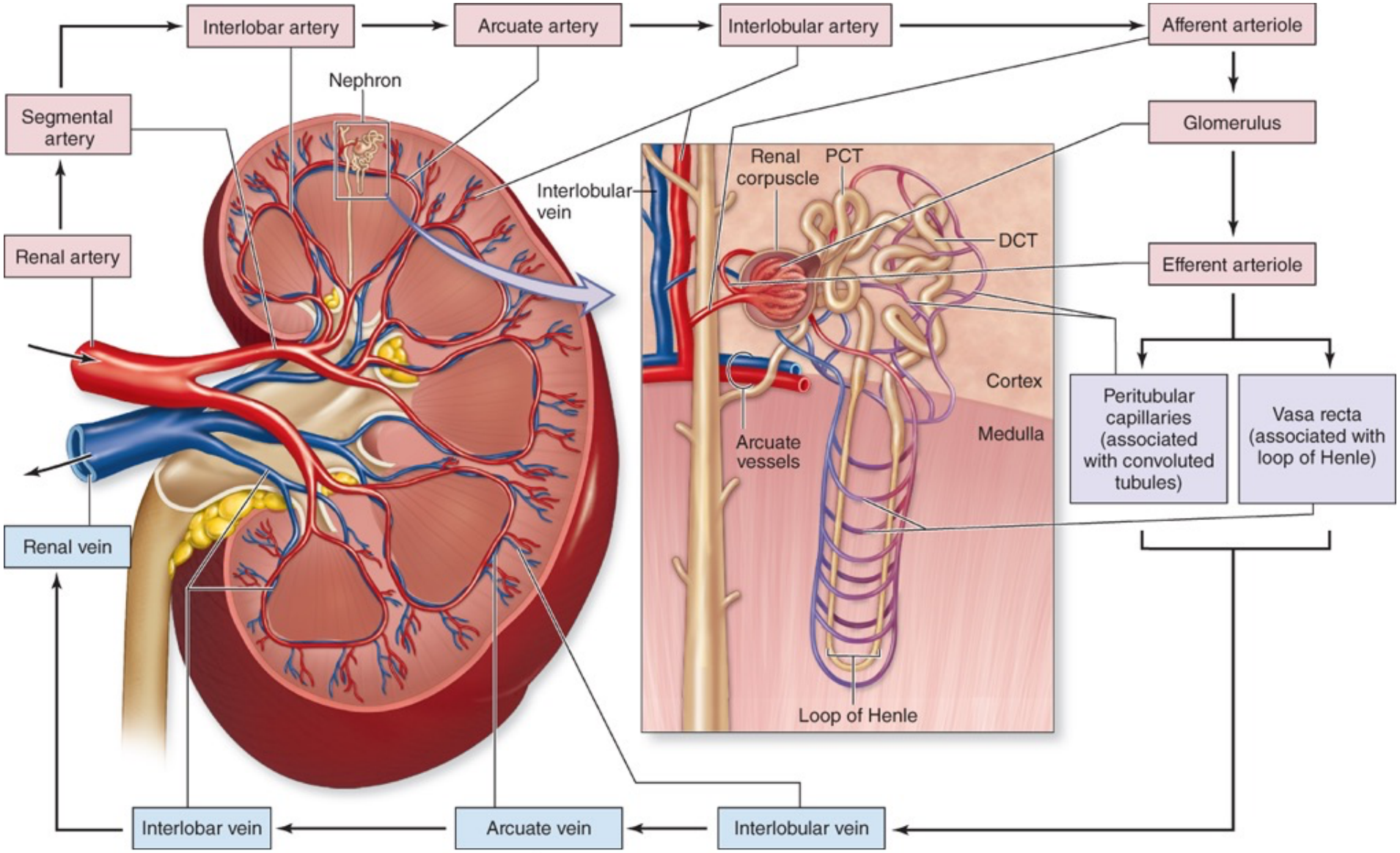
Vasa recta act as countercurrent exchanger of water & solutes that maintains the osmotic gradient of the medulla

Arterioles descending into medulla → blood loses water & gains solutes from the interstitium → hyperosmotic blood is in equilibrium with interstitium at deepest part of the loop

Venules ascending the loop → hyperosmotic blood loses salt & gains water from the interstitium

Provide circulation to tissues without disturbing the medulla's osmotic gradient

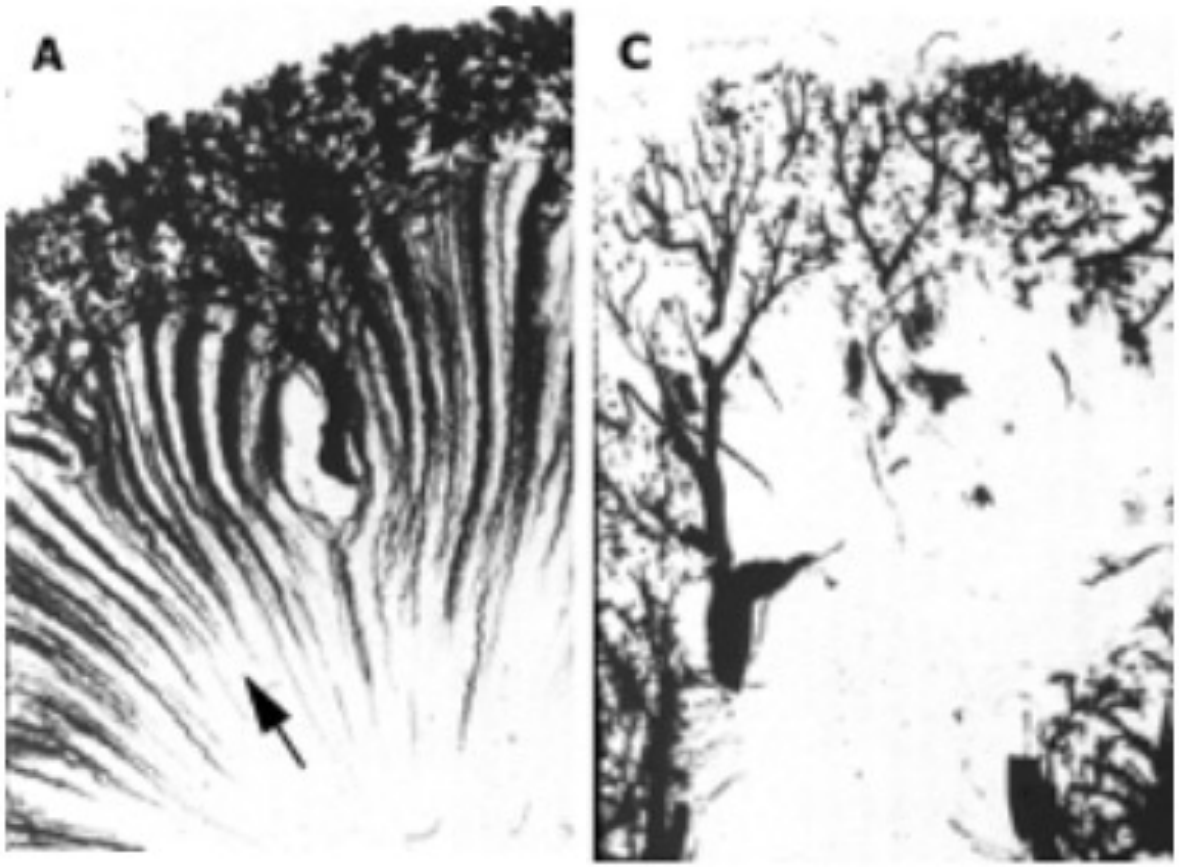
Summary of Renal Circulation



Clinical Correlate: Sickle Cell Nephropathy

- Collective term for effects of sickle cell disease on kidney function
- Affected erythrocytes sickle in the **vasa recta** (due to the hypoxic & hypertonic environment in these capillaries)
- Reduced function of the **juxtamedullary nephrons → hyposthenuria** (abnormally diluted urine)
- Can lead to medullary ischemia and infarction
- Gradual loss of glomerular & tubular function

Injection microradiographs of renal vasculature:

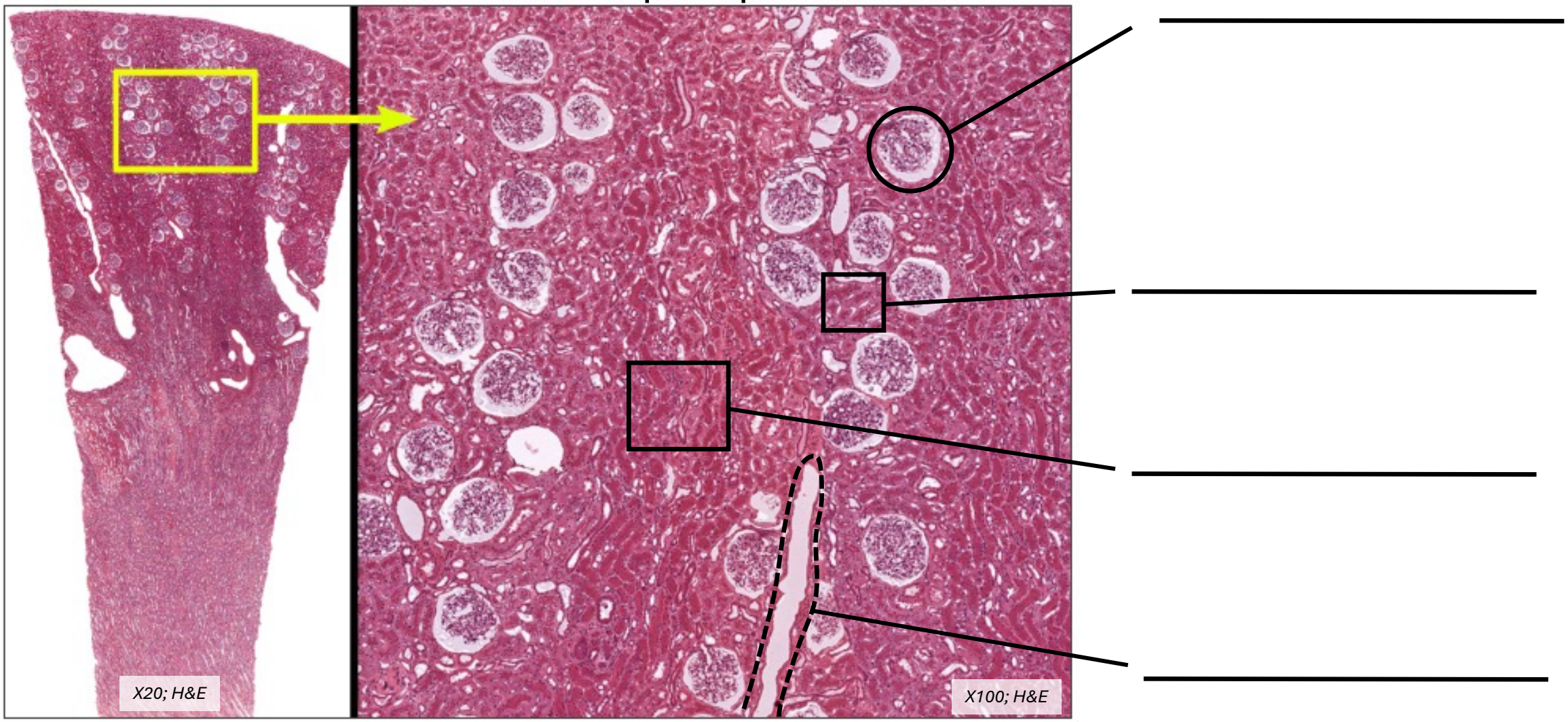


No hemoglobinopathy

Sickle cell disease

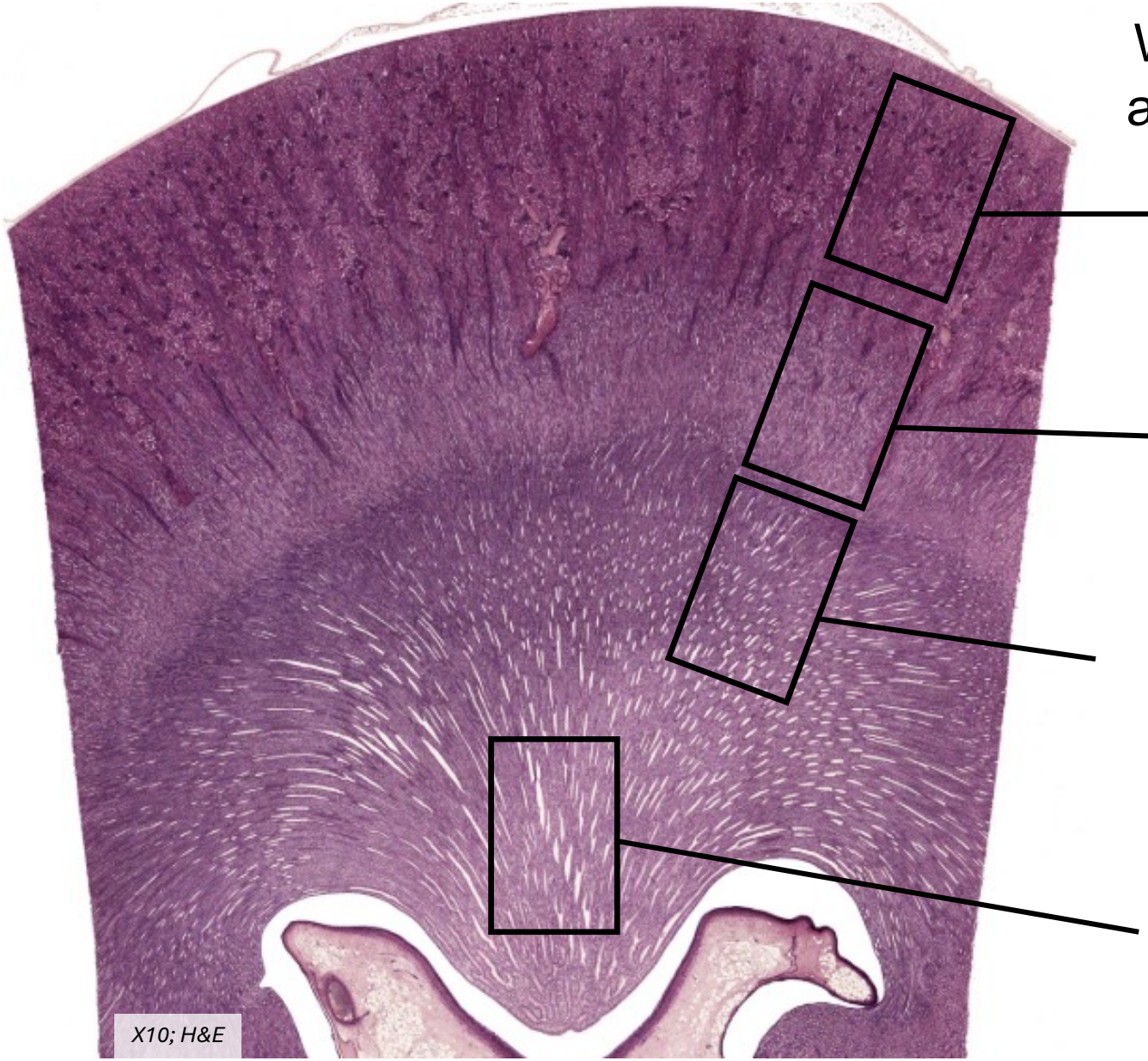
Image: UpToDate

Pause & Recall



Digital Histology

Pause & Recall

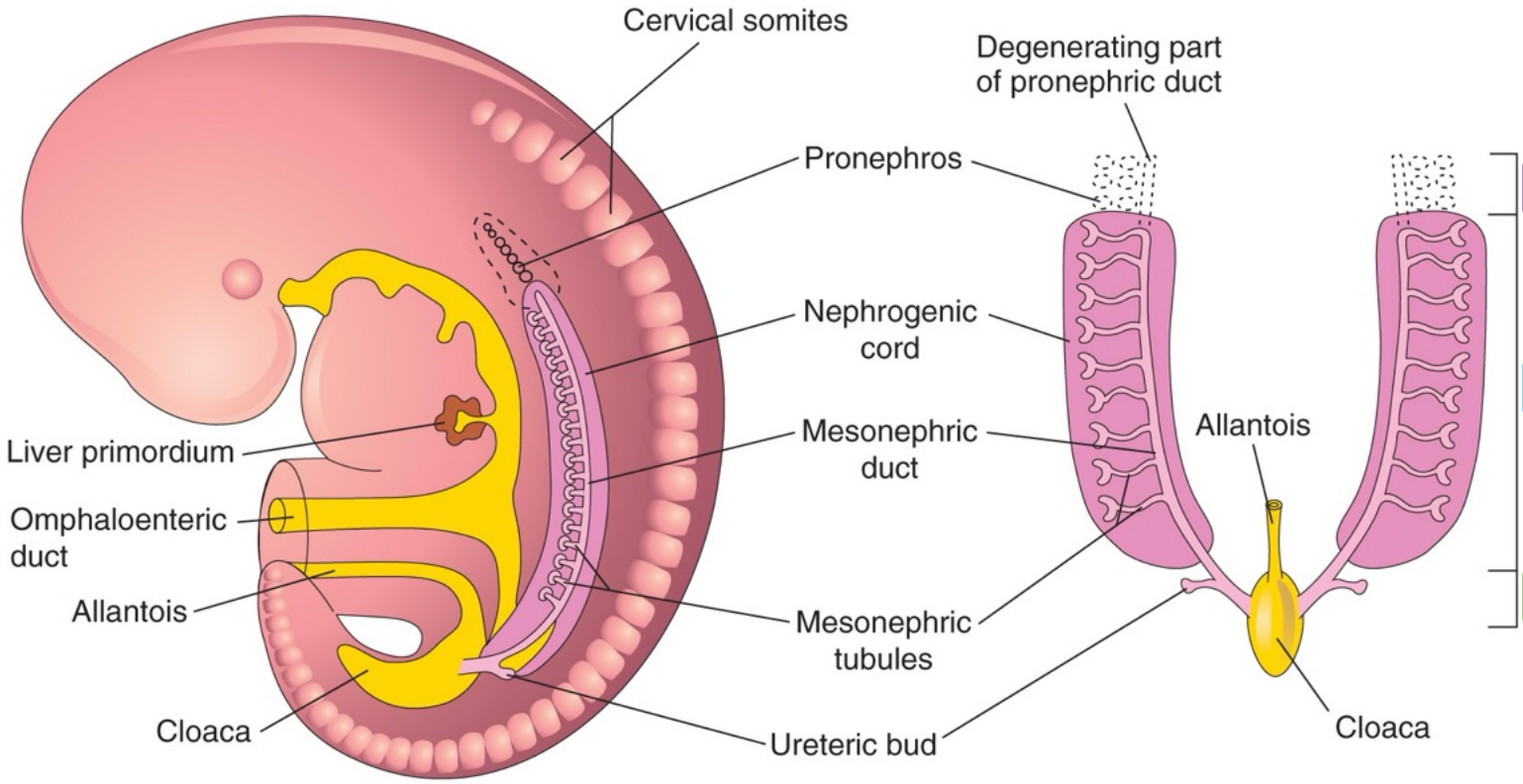


Which segments of the uriniferous tubule are found in the following kidney regions?

Digital Histology

Embryonic Kidneys

From nephrogenic cords, three sets of embryonic kidneys will form sequentially (cranial → caudal): **pronephros**, **mesonephros**, & **metanephros**



Lateral view

Ventral view

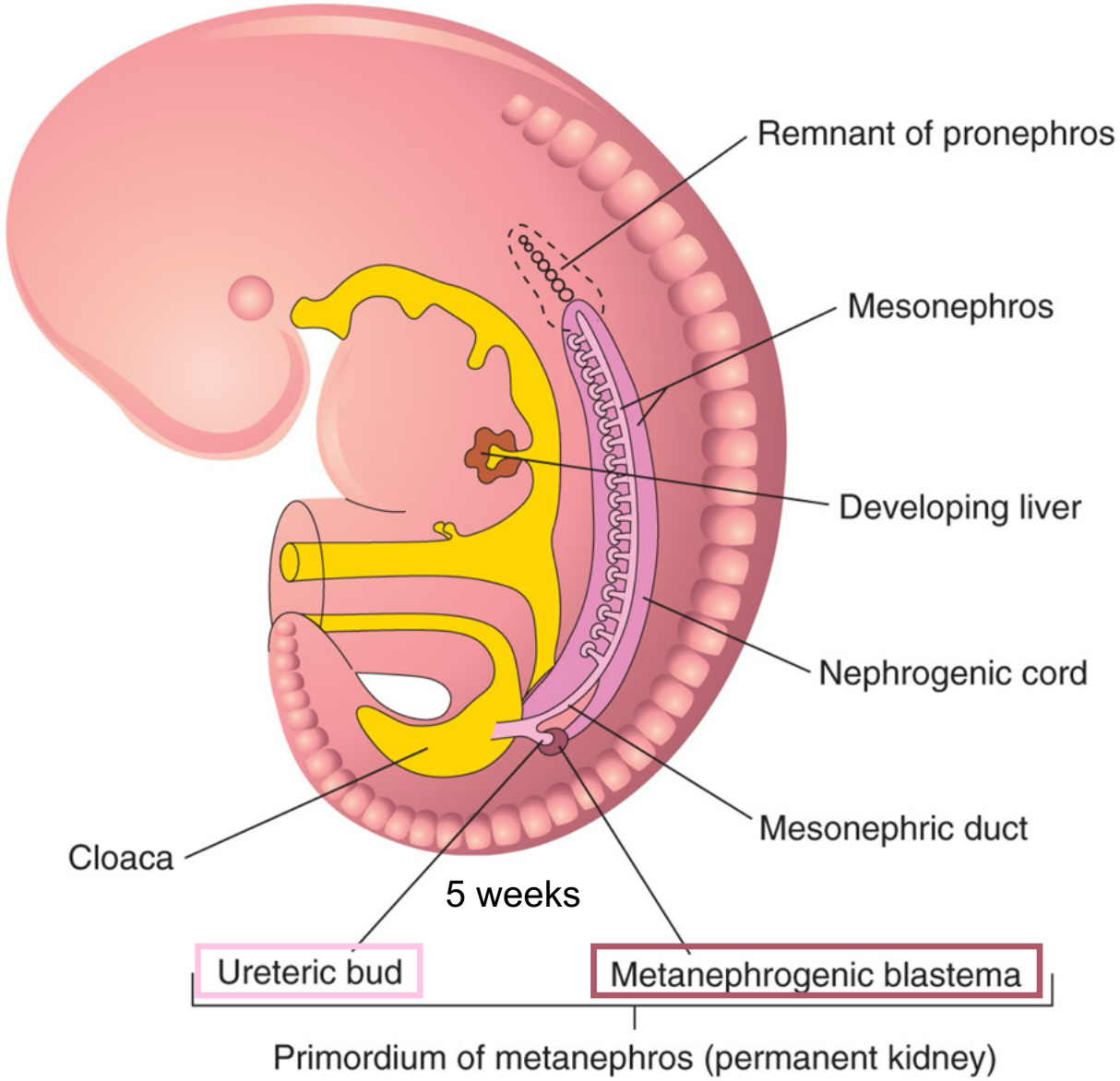
5 weeks

Pronephros → First, rudimentary set of non-functional kidneys

Mesonephros → Interim kidneys that function between Weeks 6 - 10

Metanephros → Primordia of the permanent kidneys

Metanephroi: Definitive Kidneys



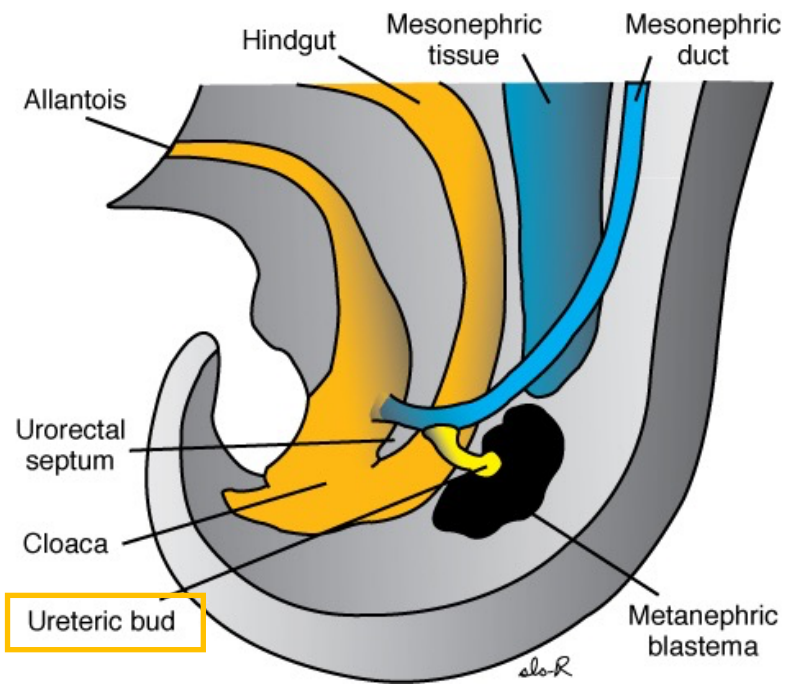
Primordia of the permanent kidney – appear in Week 5 & become functional in Week 9

Develop from two sources:

- **Ureteric bud**
  becomes collecting system
- **Metanephrogenic** (metanephric) **blastema**
  becomes excretory system

Function during fetal life to produce urine → contributes to amniotic fluid

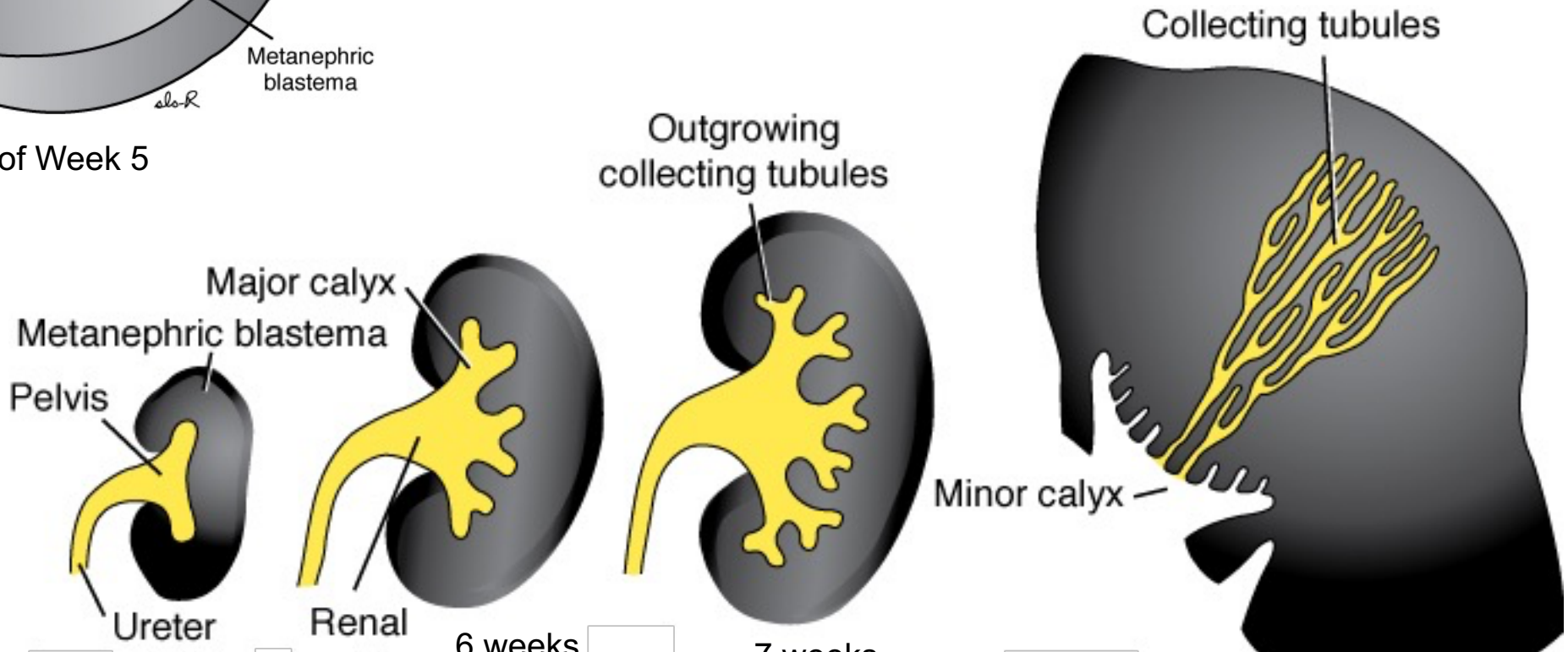
Ureteric Bud



End of Week 5

Develops as **outgrowth of mesonephric** duct near the cloaca

Projects into the metanephrogenic blastema, inducing the formation of the **collecting system**
(ureter, renal pelvis, calyces, & collecting tubules)



6 weeks

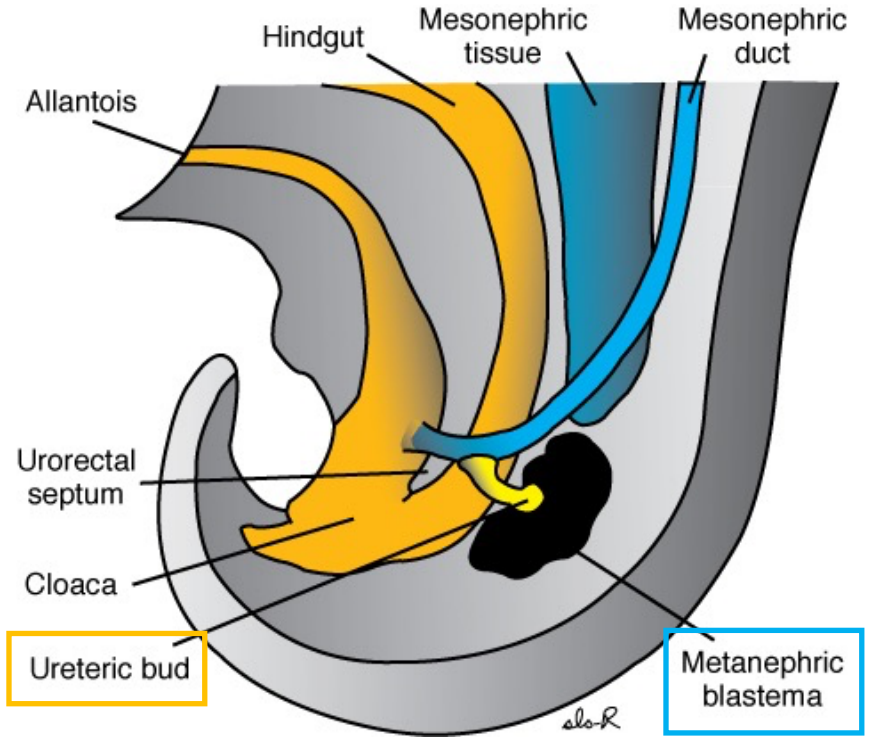
6 weeks (end)

7 weeks

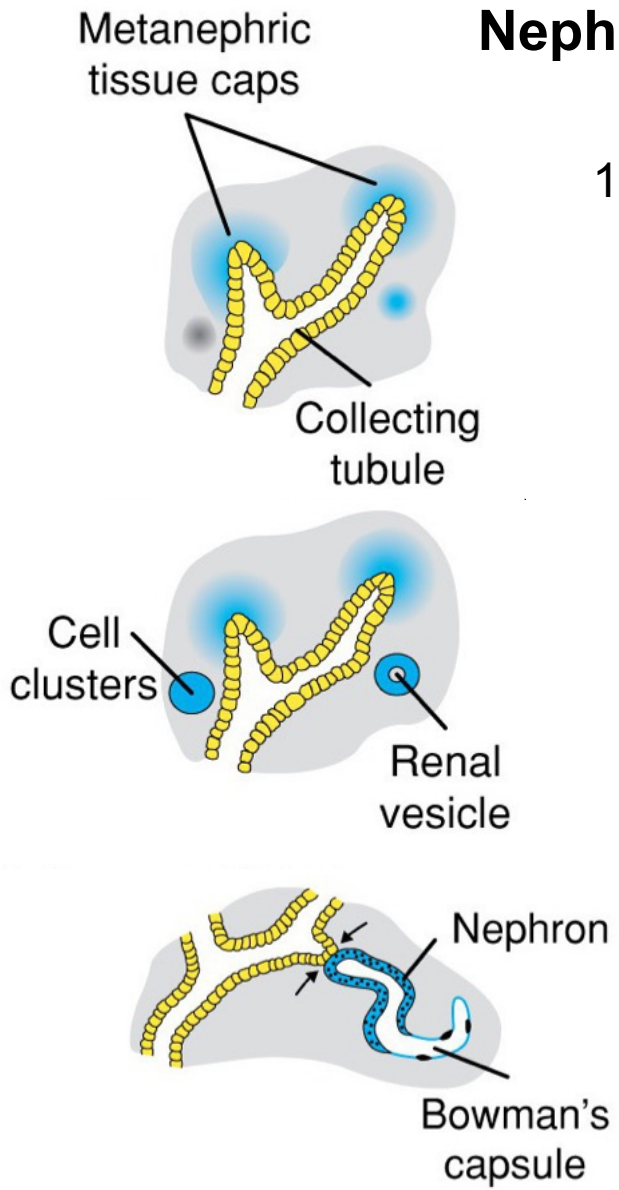
Neonate

Metanephrogenic Blastema

Develops as a **mass of metanephric mesenchyme** in the caudal part of the nephrogenic cord → gives rise to the **excretory system**



End of Week 5



Nephrogenesis begins in Week 8:

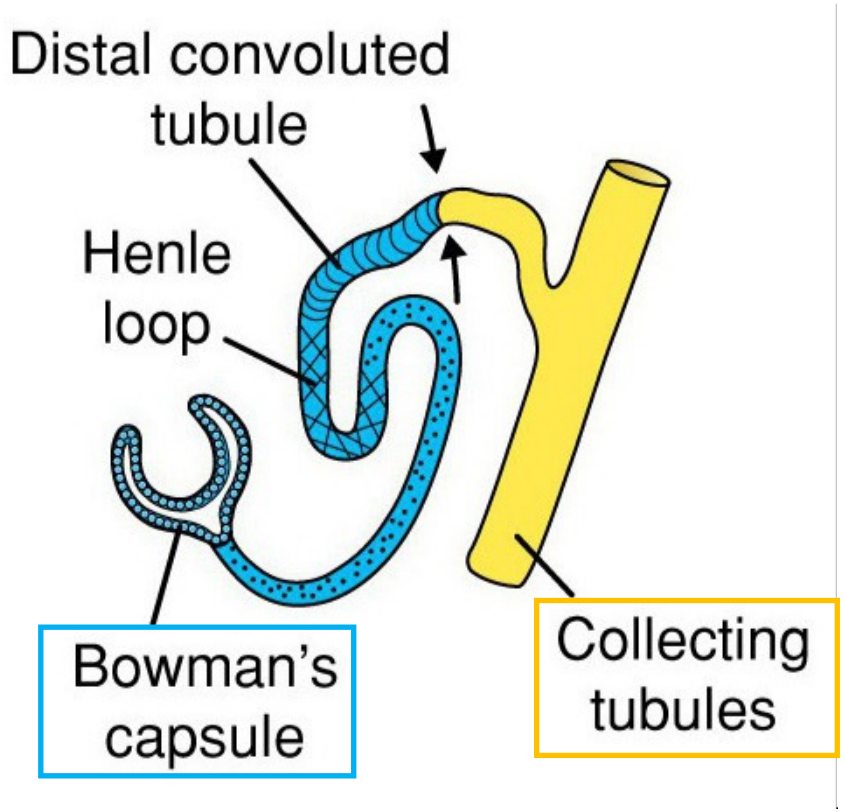
1. Newly formed collecting tubules (ureteric bud) are covered by **metanephric tissue caps** (metanephrogenic blastema)

2. Mesenchymal cell clusters in the metanephrogenic blastema form **renal vesicles**

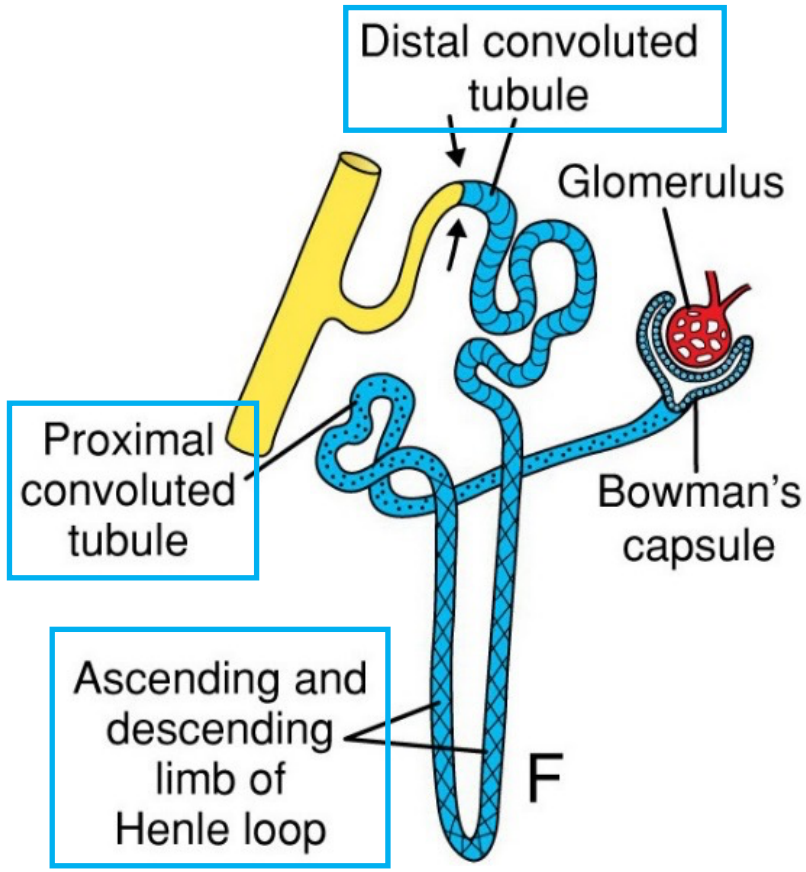
3. Vesicles develop a lumen and elongate into S-shaped **metanephric tubules**

Nephrogenesis

Proximal end of metanephric tubule forms the **Bowman's capsule**; distal end connects with the **collecting tubule** (from ureteric bud)



Metanephric tubule continues lengthening to produce:

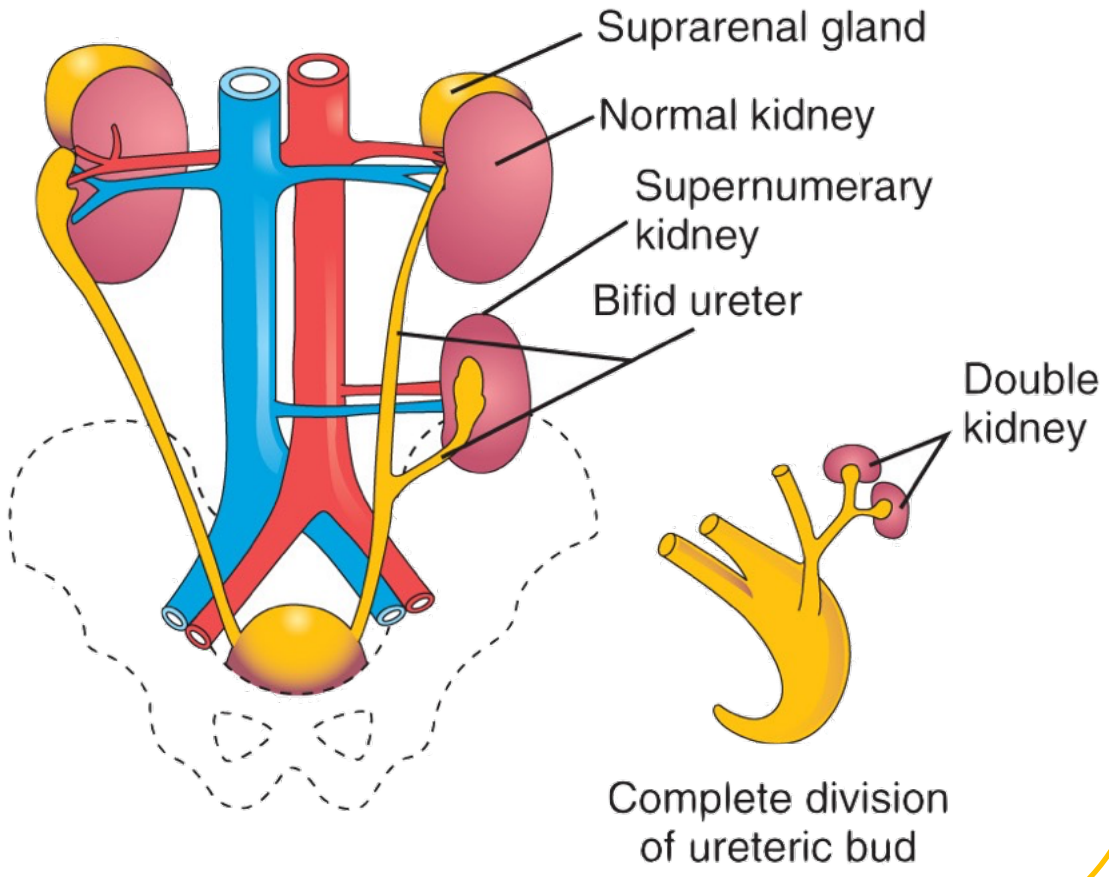


- **Proximal convoluted tubule** (PCT)
- **Nephron loop** (of Henle)
- **Distal convoluted tubule** (DCT)

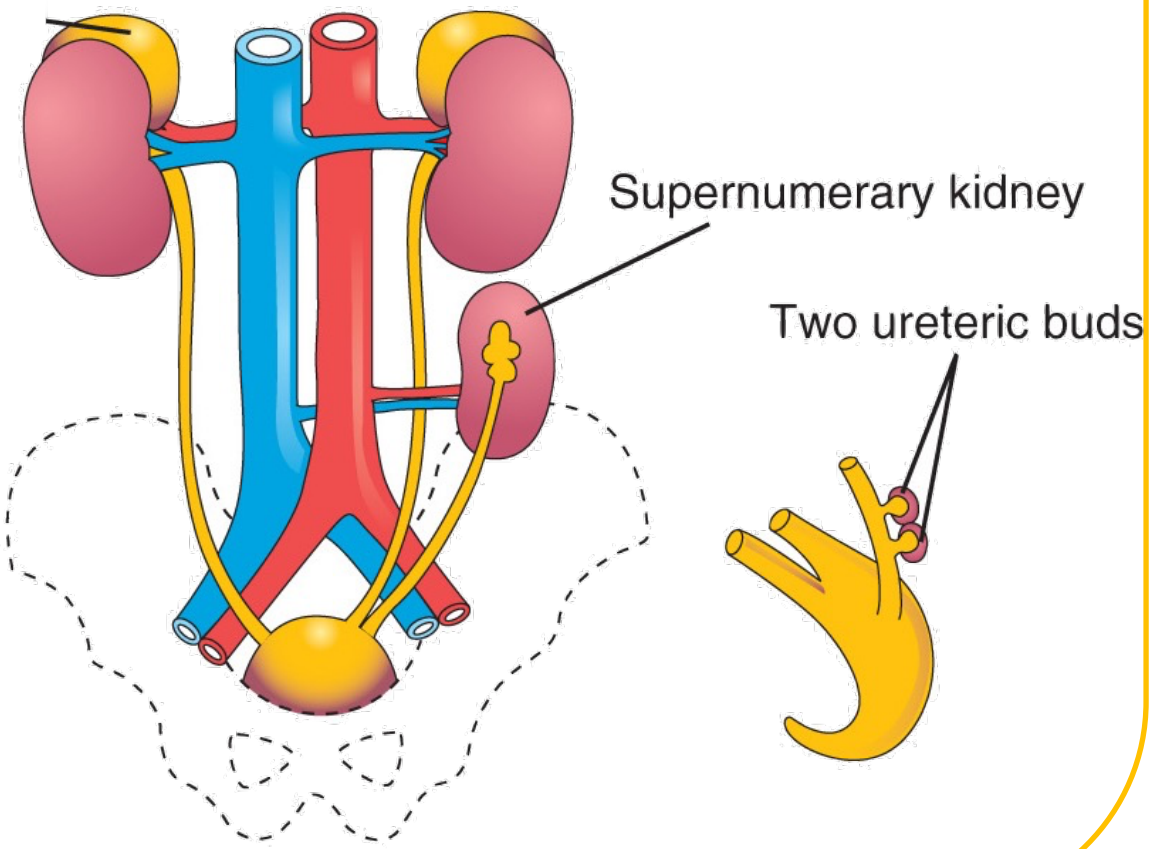
Yellow = collecting duct structures derived from the **ureteric bud**
Blue = excretory system structures derived from the **metanephrogenic blastema**

Clinical Correlate: Bifid & Duplicated Ureter

Early splitting of the ureteric bud → **bifid ureter** with single ureteric orifice at the bladder wall



Formation of a second ureteric bud → **duplicated ureter** with two ureteric orifices at the bladder wall

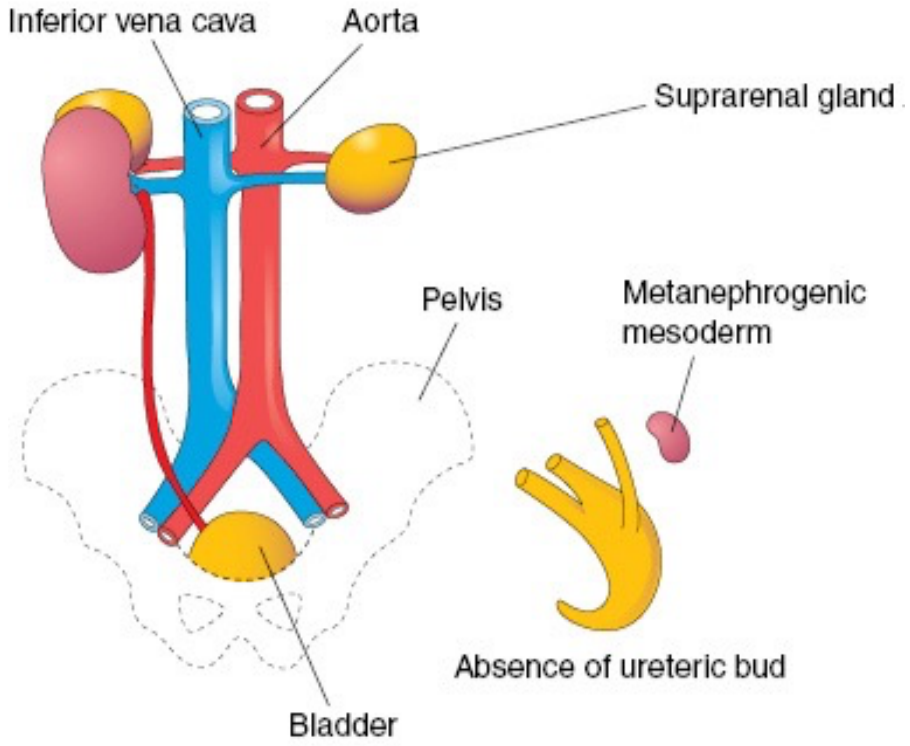


Clinical Correlate: Renal Agenesis

Results from failure of ureteric bud to form or to penetrate the metanephrogenic blastema

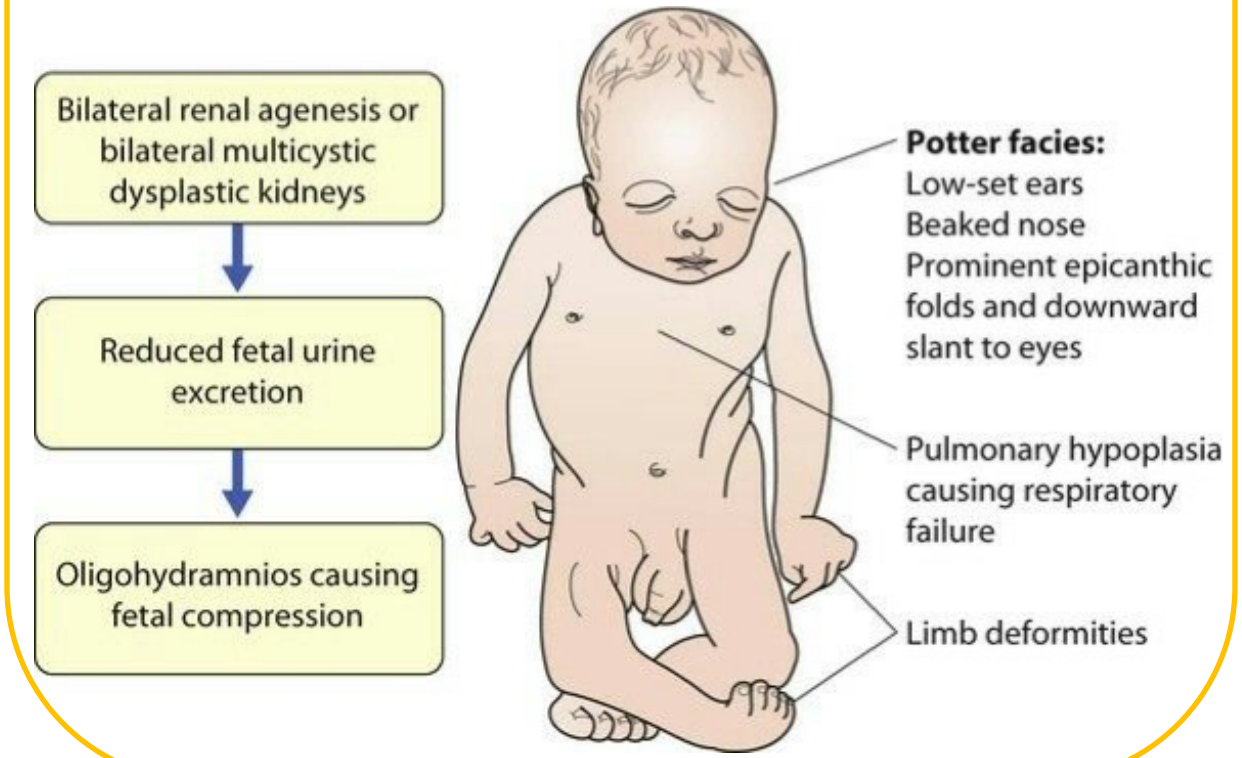
Unilateral Agenesis

- Absence of one ureteric bud
- More commonly affects left kidney
- Often asymptomatic (compensatory hypertrophy of contralateral kidney)



Bilateral Agenesis

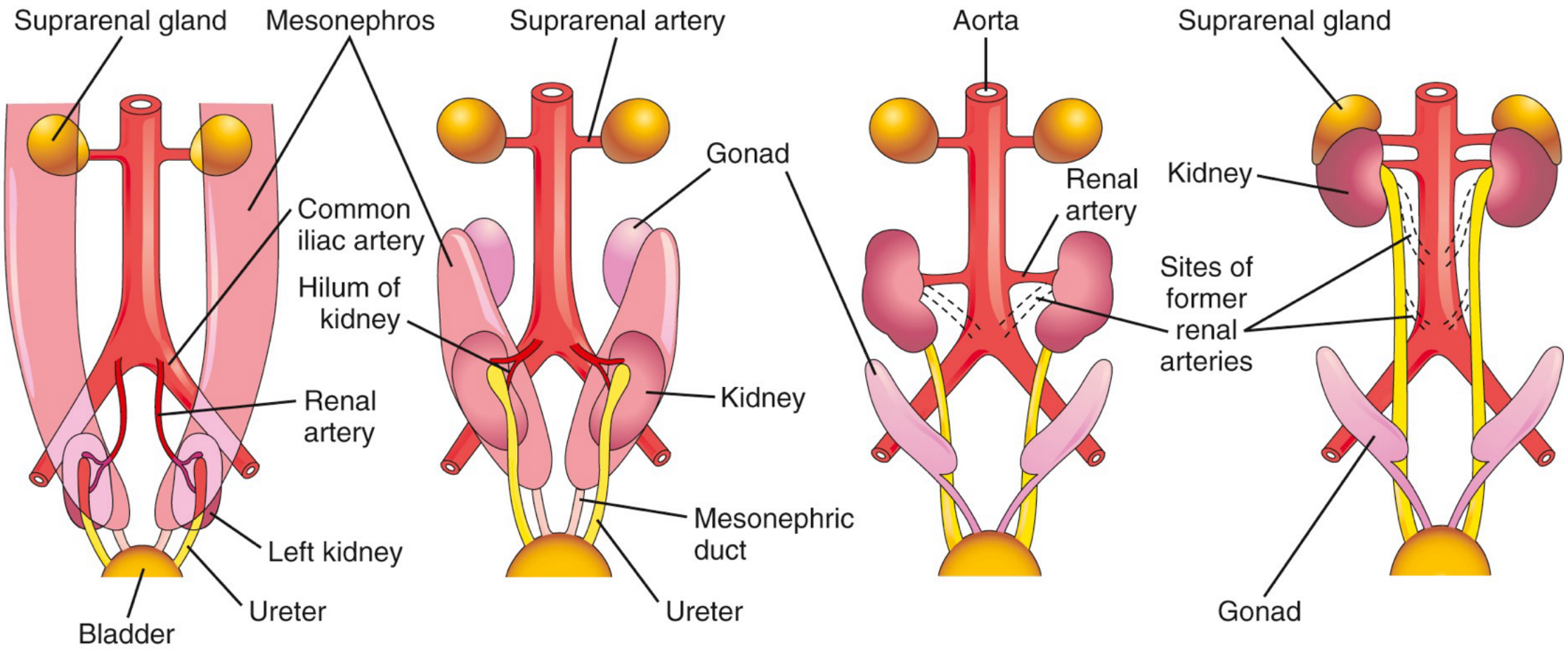
- Absence of ureteric bud on both sides
- Results in **oligohydramnios**
- Potter's phenotype, lung hypoplasia
- **Incompatible with postnatal life**



Ascent & Rotation of the Kidney

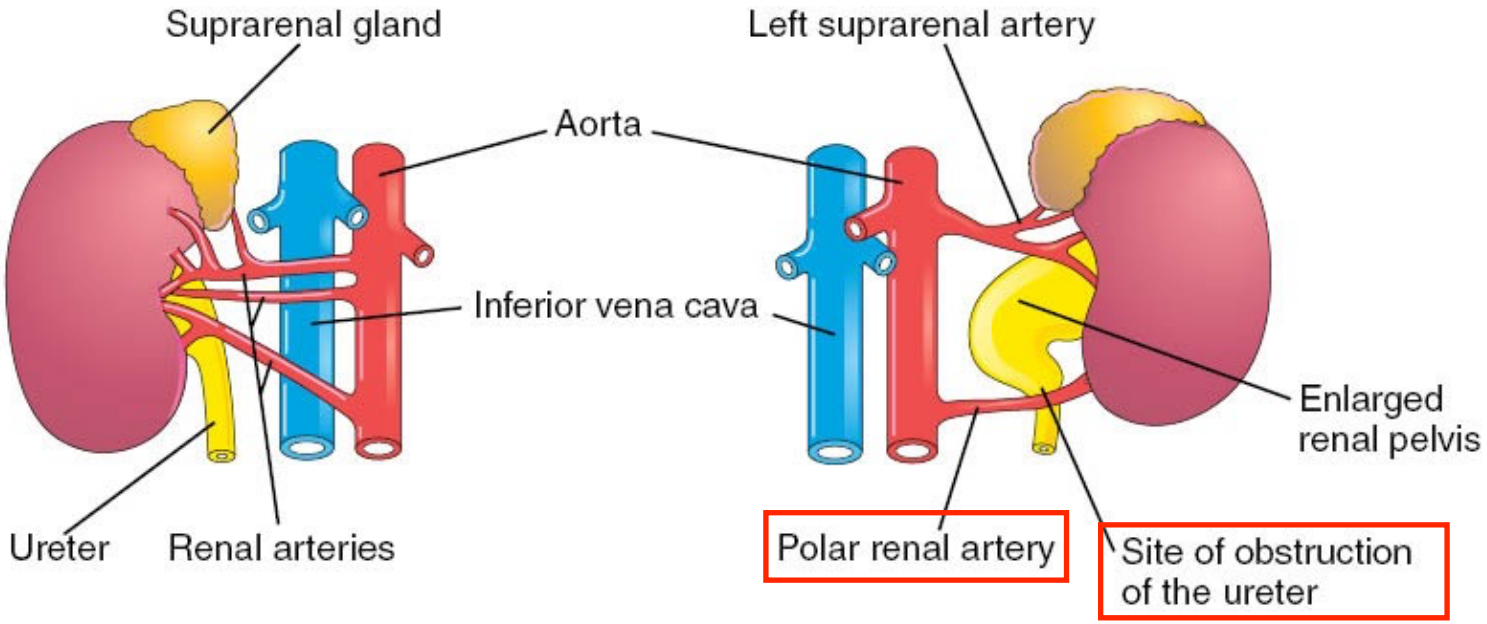
Metanephric kidneys originate in sacral region (hilum faces ventrally)

Ascend & rotate medially as trunk grows → new renal arteries form along aorta



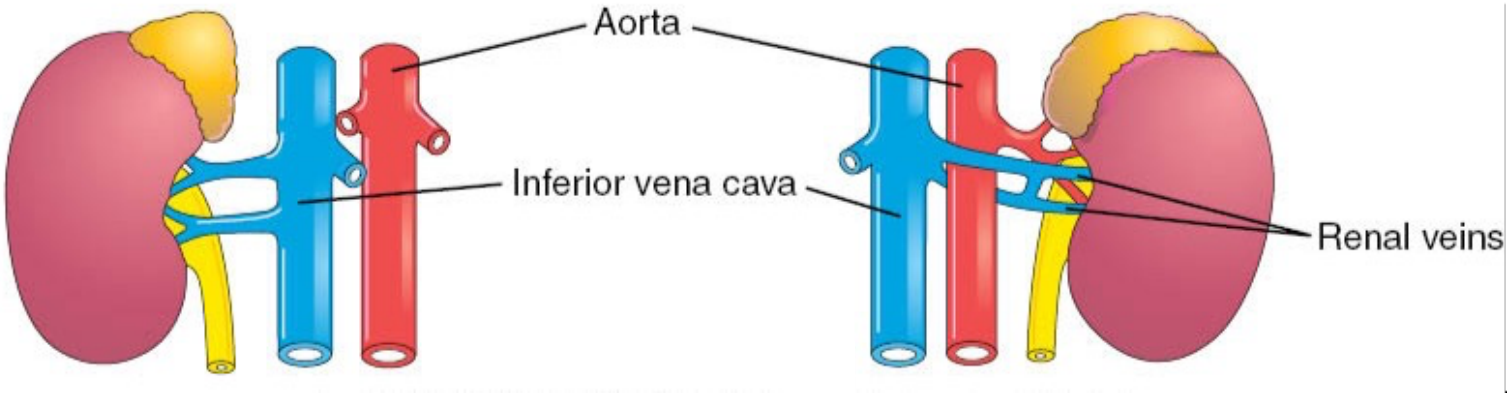
Week 6 → Week 9

Clinical Correlate: Accessory Renal Vessels



Polar renal artery

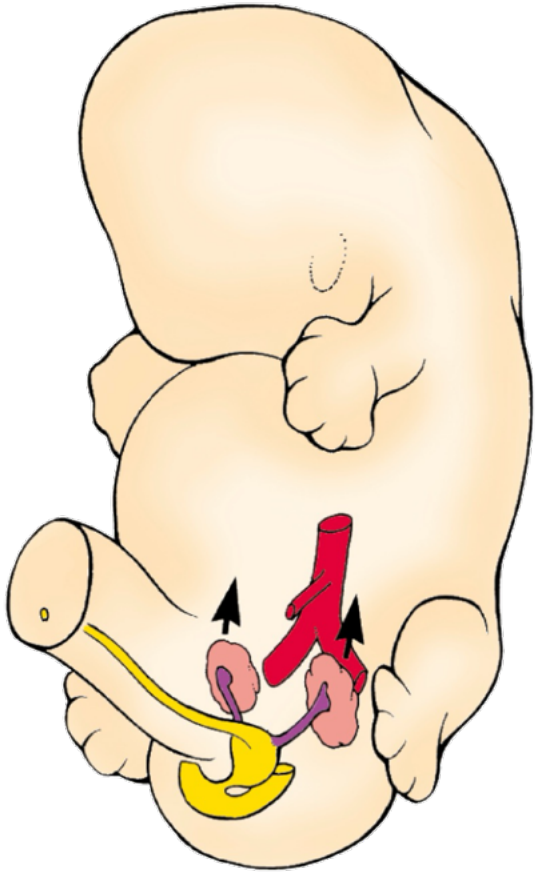
- Accessory artery entering inferior pole of the kidney
- Can **obstruct the ureter**, causing hydronephrosis



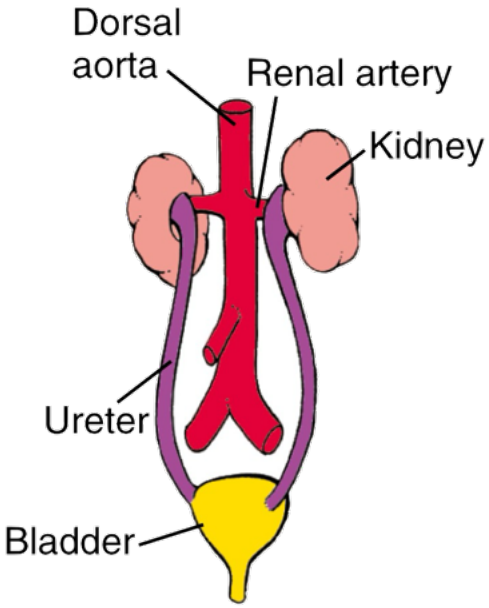
Accessory renal veins

- Half as common as accessory arteries

Clinical Correlate: Improper Ascent of Kidneys



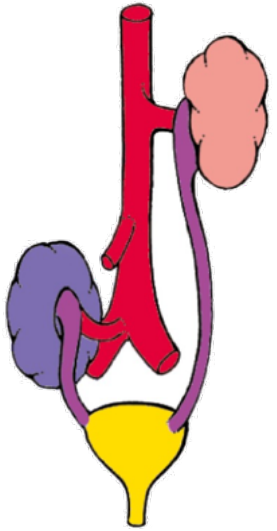
Week 6



Normal ascent of kidneys

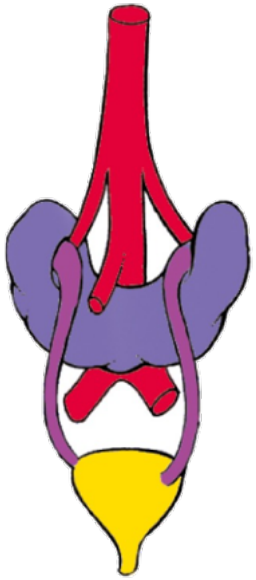
Pelvic kidney

- Failure to ascend → ectopic kidney located in the pelvis

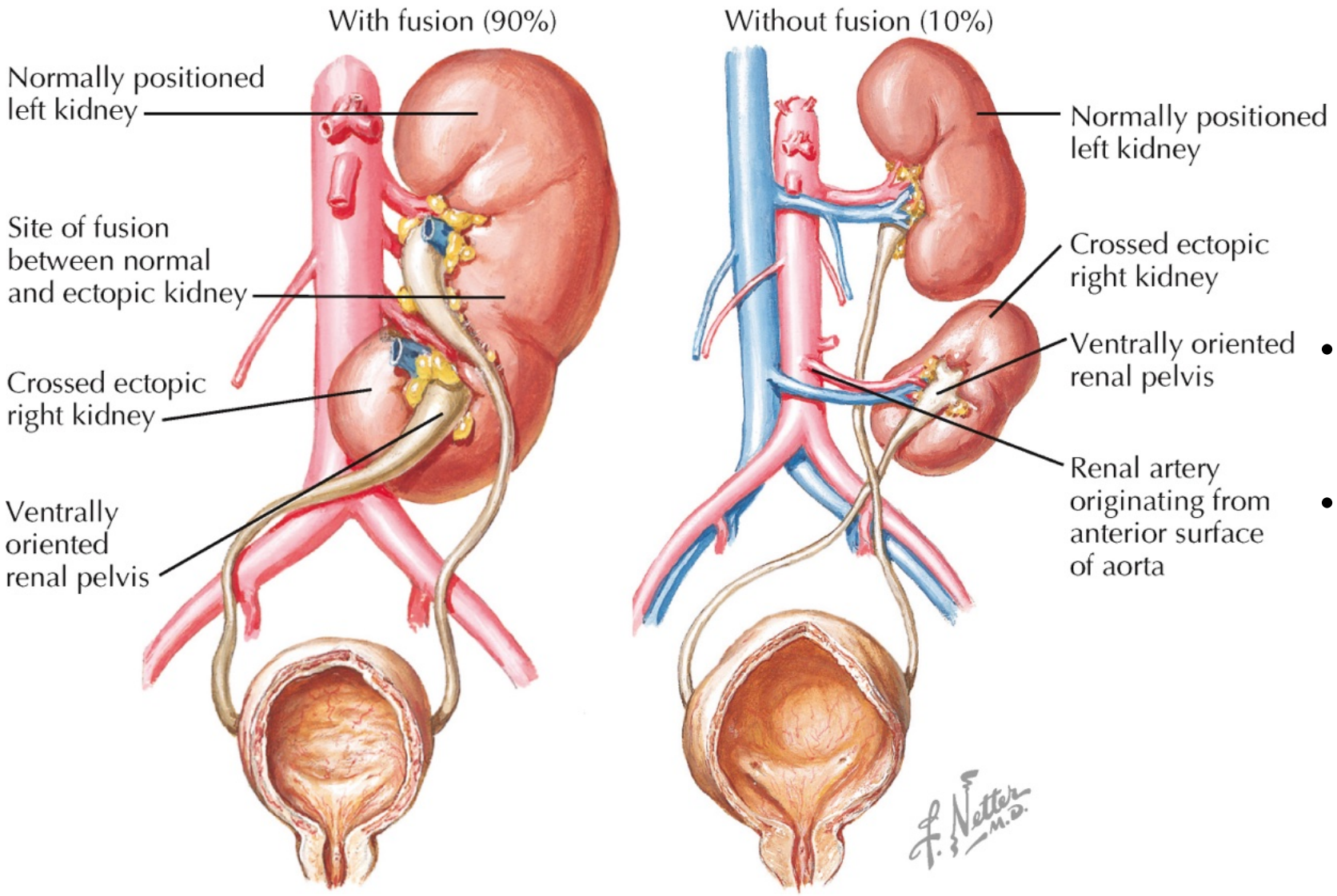


Horseshoe kidney

- Fusion of inferior poles
- Ascent is halted by the IMA
- Single U-shaped kidney in pelvis



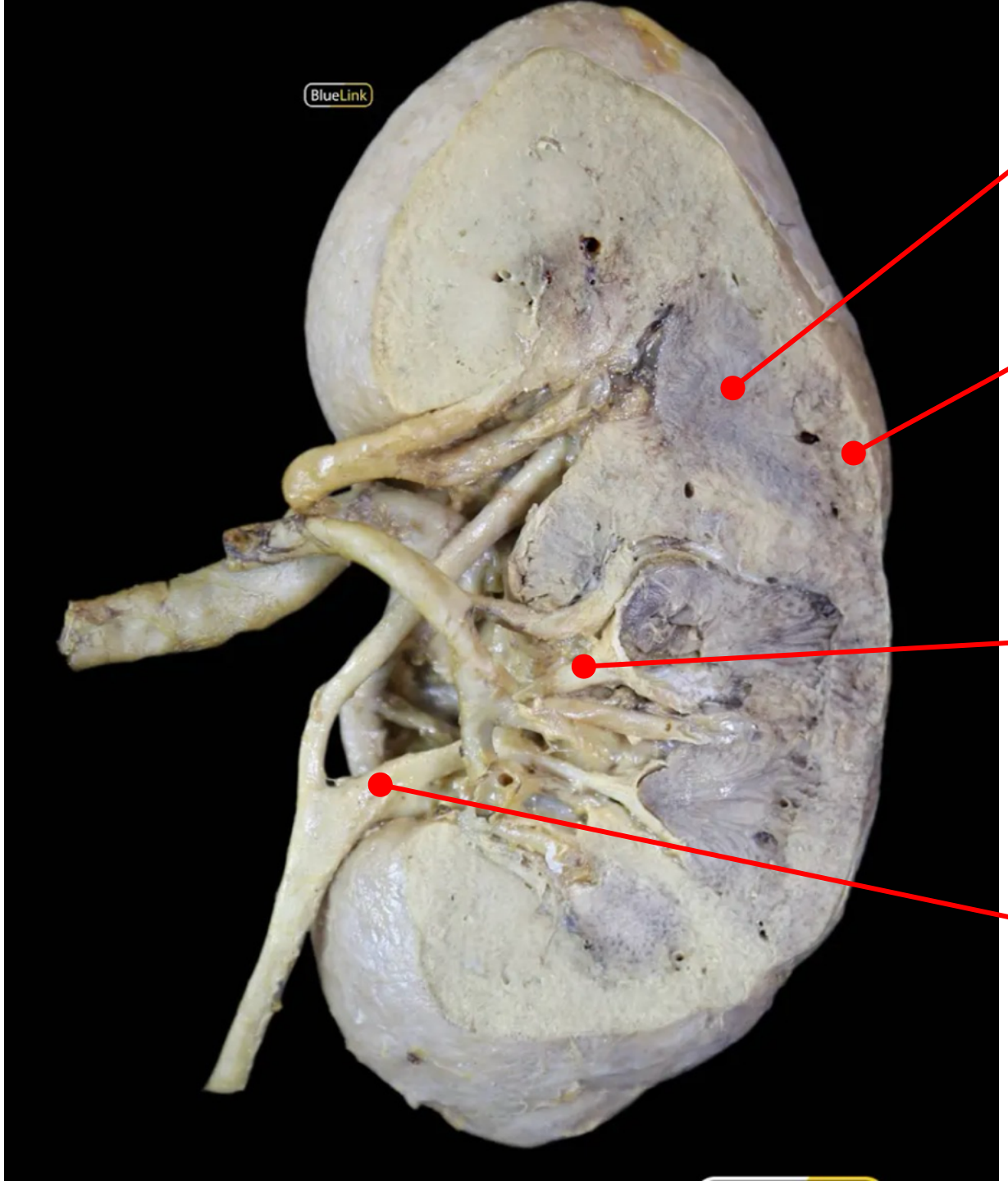
Clinical Correlate: Improper Ascent of Kidneys



Phenotypic male

Crossed renal ectopia:

- One kidney crosses midline & ascends on opposite side
- Majority of cases (90%) fuse to form a single large kidney



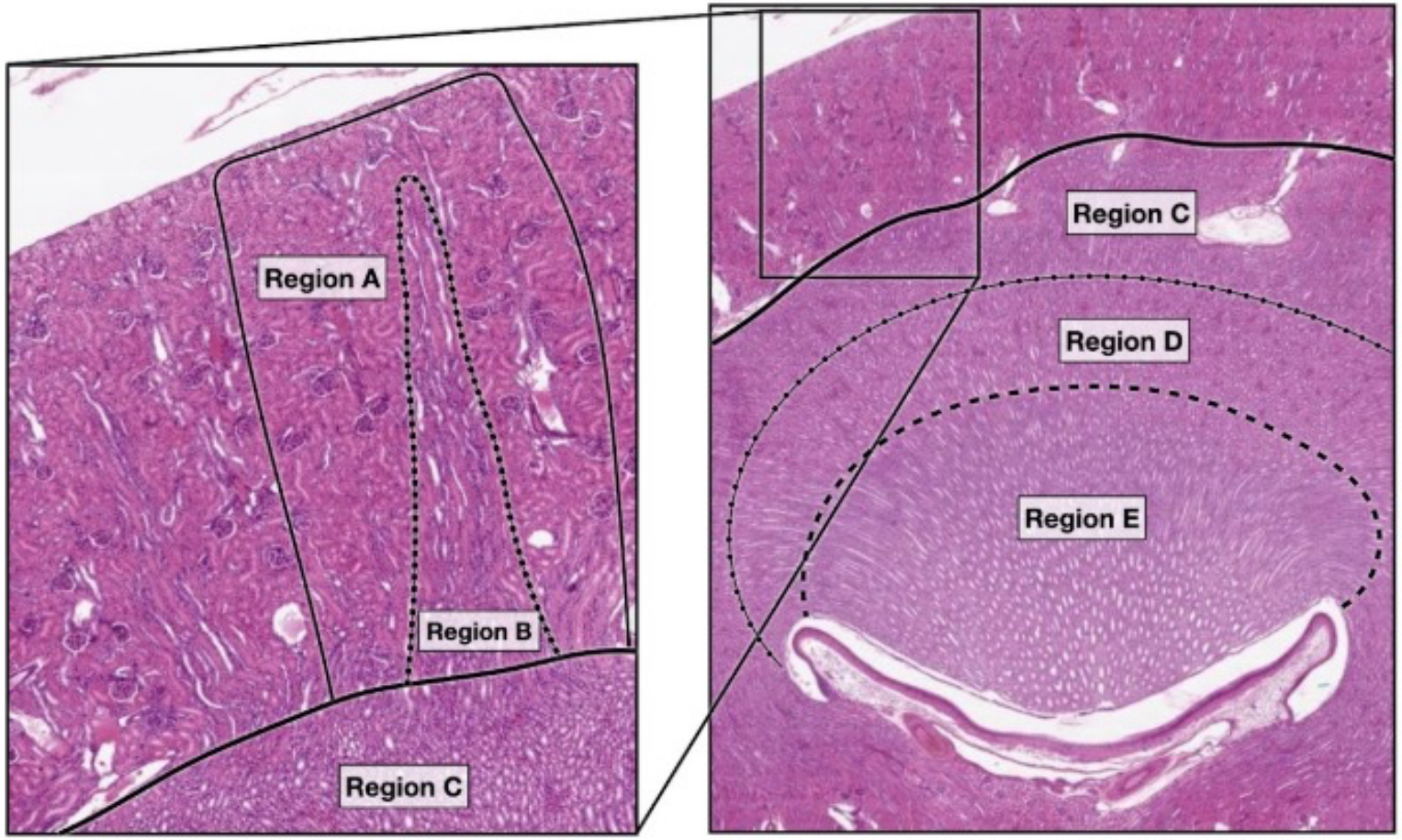
Identify the tagged structure

What embryonic structure gives rise to the nephron tubules within the tagged region?

Identify the tagged structure

What embryonic structure gives rise to tagged structure?

Session Notes & Practice Questions on the ICOM Histology Website

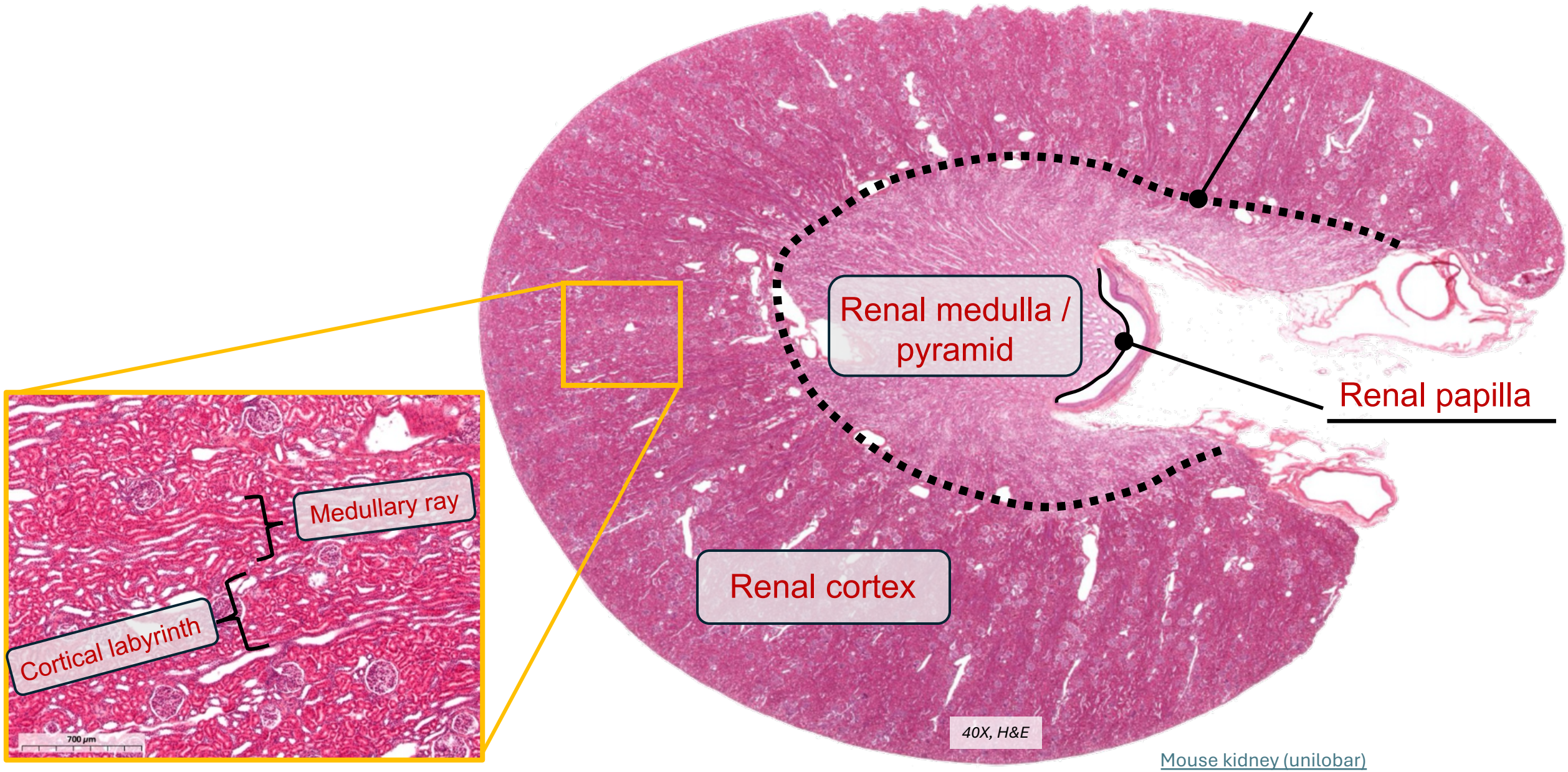


View DigitalScope Slide

In which labeled region of the kidney does filtration occur?

ANSWER KEYS

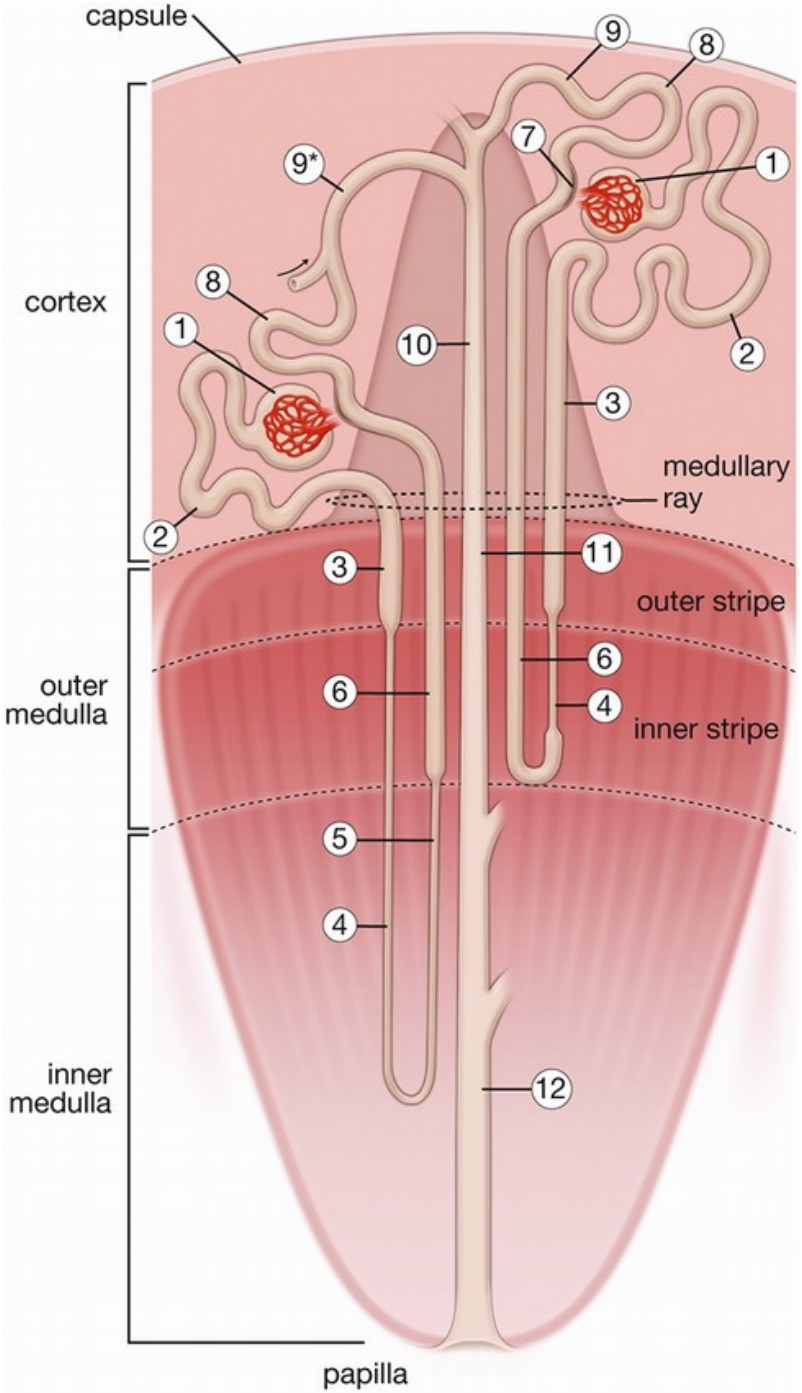
Pause & Recall



Mouse kidney (unilobar)
HistologyGuide

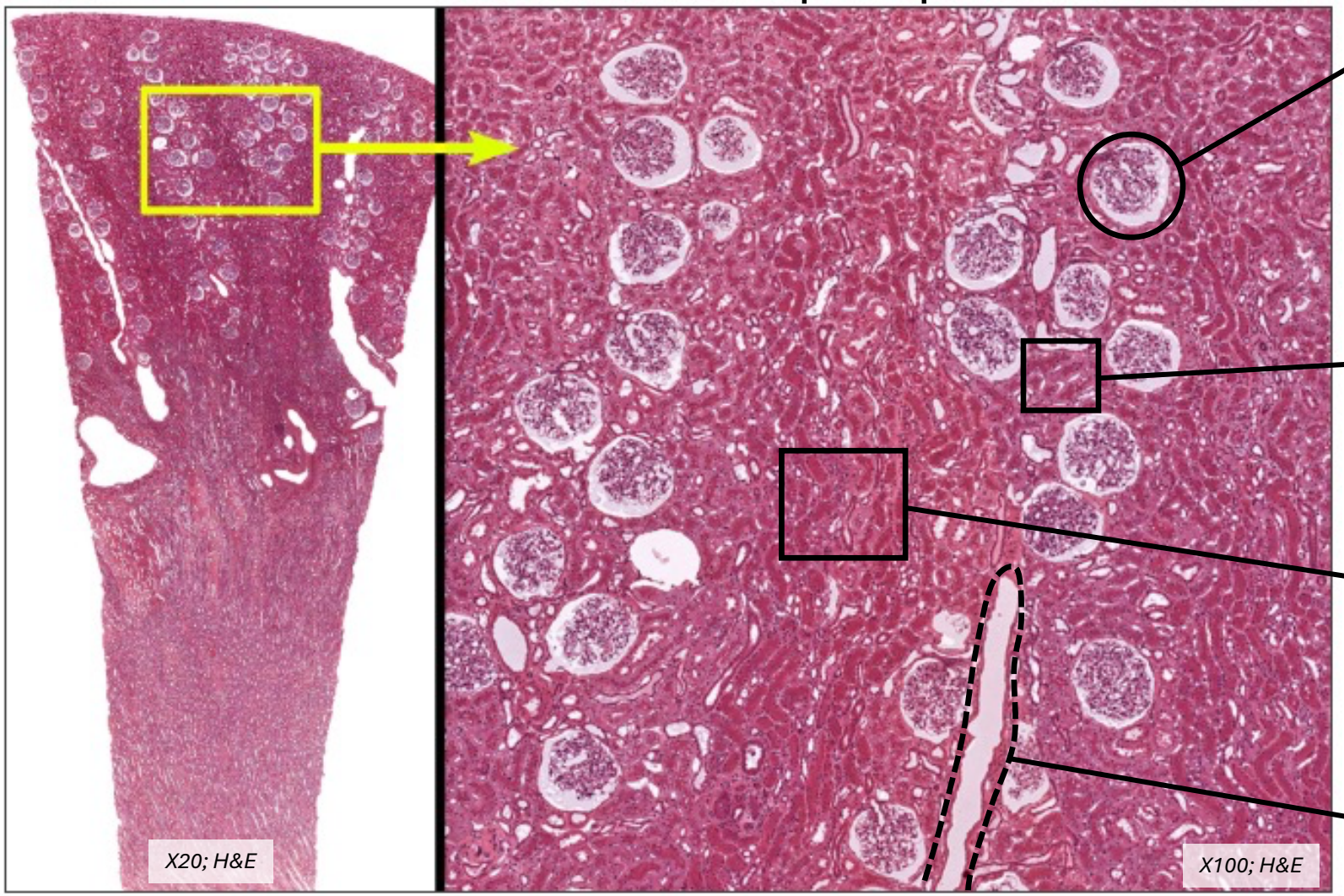
Pause & Recall

Identify the numbered components of the uriniferous tubule



1. Glomerular (Bowman's) capsule
2. Proximal convoluted tubule
3. Thick descending limb (proximal straight tubule)
4. Thin descending limb
5. Thin ascending limb
6. Thick ascending limb (distal straight tubule)
7. Macula densa
8. Distal convoluted tubule
9. Connecting tubule
10. – 12. Collecting duct

Pause & Recall



Cortical labyrinth

Medullary ray

Renal corpuscle

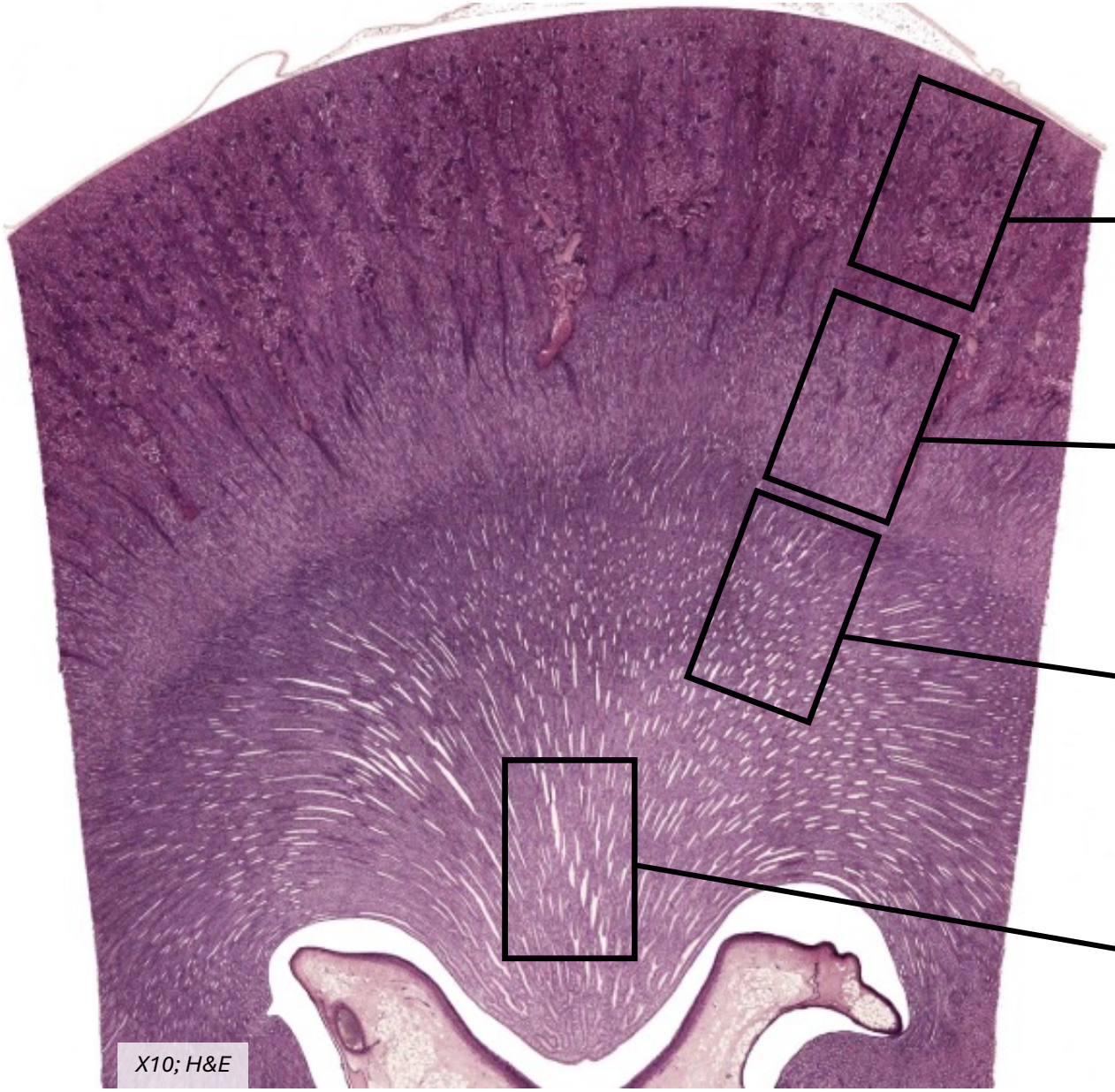
Convoluted tubules

Collecting duct & straight tubules

Interlobular artery

Digital Histology

Pause & Recall



Digital Histology

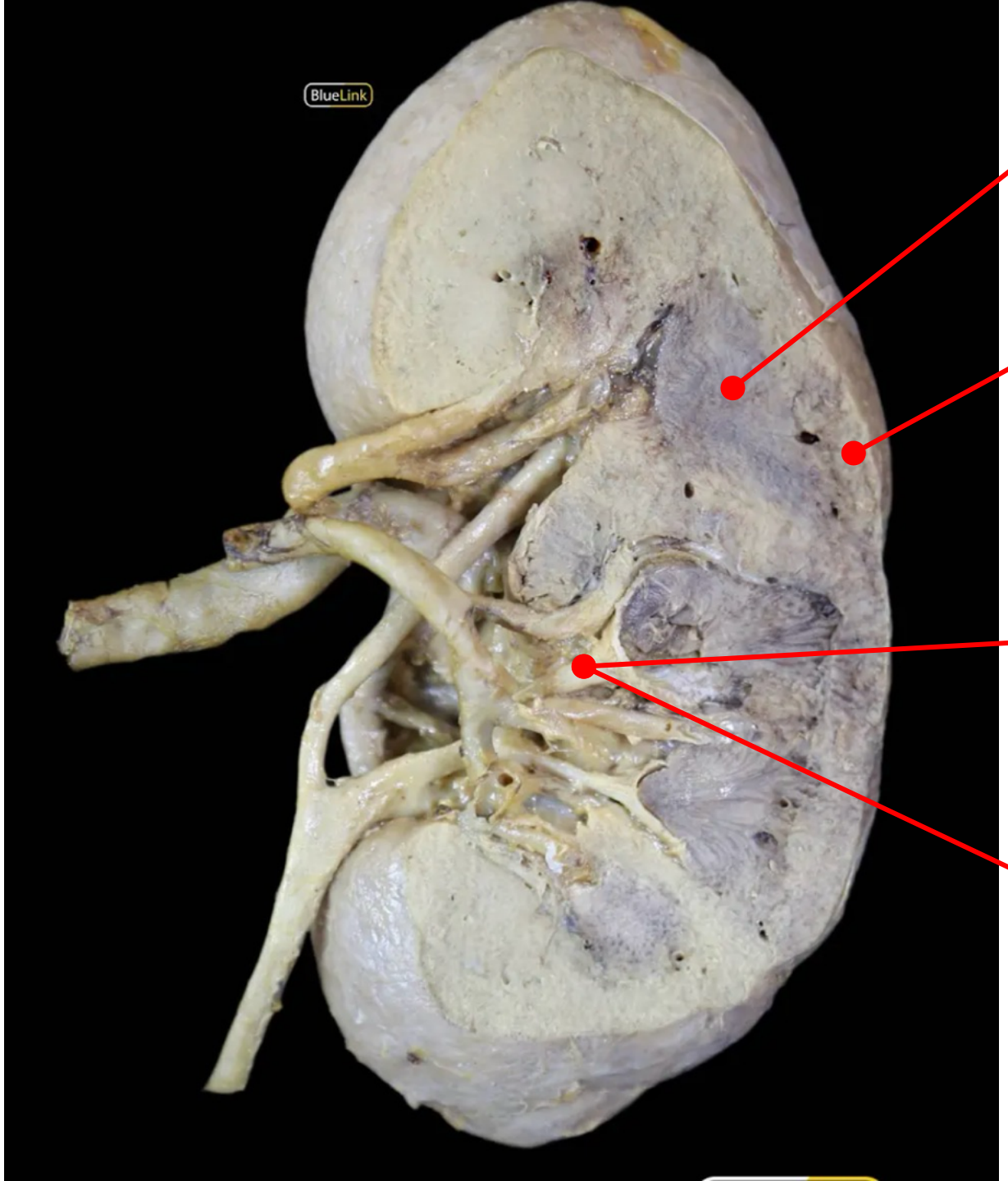
Which segments of the uriniferous tubule are found in the following kidney regions?

Renal Cortex:
- Renal corpuscle, PCT, DCT (cortical labyrinth)
- Collecting duct, PST, DST (medullary ray)

OSOM:
- Thick descending limb / PST & thick ascending limb / DST
- Collecting duct

ISOM:
- Thin descending limb & thick ascending limb / DST
- Collecting duct

Inner Medulla:
- Thin descending limb & thin ascending limb
- Collecting duct → papillary ducts



Identify the tagged structure

Renal pyramid

What embryonic structure gives rise to the nephron tubules within the tagged region?

Metanephrogenic blastema

Identify the tagged structure

Minor calyx

What embryonic structure gives rise to tagged structure?

Ureteric bud